Language-Inspired Hierarchical Mamba Network With Superpixel-Structured Training for Hyperspectral Image Classification

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***Abstract*—Existing Mamba-based hyperspectral image classification (HSIC) methods typically decouple spectral and spatial modeling into two independent and parallel branches and fuse their features afterward. Such physical separation risks information loss or feature conflicts when the fusion strategy is suboptimal, ultimately degrading the classification accuracy. To address this issue, we propose a language-inspired hierarchical Mamba network with superpixel-structured training (L-HMambaS) that unifies the spectral and spatial dimensions from the outset. Specifically, inspired by the structure of natural language, we regard the spectrum of a pixel as a "word" and the spatial arrangement of multiple pixels as a "sentence" and devise a nested Mamba architecture. In the L-HMambaS model, an inner spectral Mamba [inner word-level mamba (ISM)] block captures cross-band dependencies among pixels, while an outer spatial Mamba (OSM) block explores local contextual relations among spatial patches. Furthermore, to suppress redundancy induced by overlapping patches in conventional sampling strategies, we introduce a superpixel-driven sampling strategy that guarantees high intrapatch homogeneity and minimal spatial overlap. Extensive experiments on five publicly available hyperspectral datasets, namely, Pavia University, WHU-Hi-LongKou, WHU-Hi-HanChuan, WHU-Hi-HongHu, and Houston 2013, demonstrate that L-HMambaS achieves overall accuracies of 97.01%, 98.17%, 92.91%, 93.37%, and 96.79% with only 30 labeled samples per class, consistently surpassing state-of-the-art competitors. The codes will be available from the website: https://github.com/rachelitem/L-HMambaS**

***Index Terms*—Hyperspectral image classification (HSIC), Mamba, spectral–spatial feature, superpixel.**

I. Introduction

With the continuous development of Earth observation technologies, hyperspectral imaging (HSI) has emerged as a prominent research area in both remote sensing and computer vision. HSI captures reflectance information of ground objects across tens to hundreds of contiguous spectral bands, spanning the visible to near-infrared and even mid- and far-infrared regions. This enables each pixel to be represented by a high-dimensional, continuous spectral signature [1]. The rich and fine-grained spectral information allows HSI to effectively distinguish materials that are spectrally similar in conventional RGB imagery [2], offering reliable data support for a wide range of applications, including material identification [3], environmental monitoring [4], agricultural assessment [5], resource exploration [6], and military reconnaissance [7]. Among these, hyperspectral image classification (HSIC), which aims to assign a unique land-cover label to each pixel, is a fundamental task in hyperspectral remote sensing.

In recent years, deep learning has significantly advanced HSIC. Among, CNN-based approaches [8], [9], [10], [11], [12] excel in extracting local spectral–spatial features, whereas graph convolutional networks (GCNs) [13] effectively encode pixel-level relations; nevertheless, both paradigms struggle to capture long-range dependencies [14]. Transformer architectures [15], [16], [17], [18] achieve global spectral–spatial modeling via self-attention, yet their quadratic complexity constrains scalability [19]. Hybrid CNN–Transformer designs [20], [21] balance local and global representations, but still inherit the inherent limitations of each component [22].

To mitigate these burdens, recent research has pursued models that retain global-receptive-field modeling at linear cost. The Mamba architecture, grounded in a state-space model (SSM) [23], realizes this goal by compressing long sequences into the hidden state space that evolves through parameterized state transitions and selective gating mechanisms [24], yielding linear computational complexity while preserving long-range dependencies. Consequently, Mamba dispenses with the quadratic-complexity attention operations of Transformers and has exhibited efficient inference performance in natural language processing, speech modeling, and computer vision [25], also covering HSIC. MambaHSI [26] stacks multiple cascaded Mamba blocks, where each block consists of parallel spectral and spatial branches to extract complementary features. IGroupSS [27] employs an intergroup selective state-space (IGSM) mechanism, which splits the spectral dimension into four regular, nonoverlapping groups, and applies unidirectional S6 operators along each

Received 22 January 2026; revised 2 April 2026 and 22 April 2026; accepted 20 May 2026. Date of publication 22 May 2026; date of current version 1 June 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62271130 and Grant T2225019, in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2023A1515010066, and in part by the Science and Technology Foundation of Guangdong Province under Grant 2021A0101180005. *(Corresponding author: Xiang Xu.)*

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Digital Object Identifier 10.1109/TGRS.2026.3696076

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group, pruning spectral redundancy before global modeling. HSIMFormer [28] hybridizes intrascale transformer experts (ITEs) with cross-scale Mamba experts (CMEs) to exploit the complementary advantages of Transformer and Mamba architectures. Collectively, these designs confirm that Mamba architecture can deliver both representation power and computational thrift for high-dimensional hyperspectral data.

On the other hand, considering that Mamba [23] was originally designed for causal sequence modeling in natural language processing, its 1-D and unidirectional scanning mode [29] is incompatible with the noncausal, spatially structured visual images. To address this, VMamba [30] introduces a cross-scan module (CSM) that performs four-directional scanning, enabling each spatial pixel to aggregate contextual information from all directions and thus yielding a global receptive field. When extending Mamba to HSIC, the scanning mechanism becomes even more critical. Hyperspectral data are characterized by complex 3-D spectral–spatial (3DSS) correlations, requiring more sophisticated scanning mechanisms beyond spatial grid traversal. S2Mamba [31] proposes a block-wise cross-scanning strategy to capture spatial correlations among pixels, complemented by a bidirectional spectral scanning strategy to capture interband relationships. Furthermore, 3DSS-Mamba [32] incorporates a 3DSS selective scanning mechanism, enabling pixel-wise modeling within unified 3DSS representations.

It is worth noting that existing Mamba-based HSIC methods primarily rely on dual-branch architecture to separately extract spectral and spatial features [26], [33] or, as in 3DSS-Mamba [32], employ specific scanning strategies to preprocess hyperspectral data. After feature processing, the spectral signature of a pixel exhibits strong similarity to word embeddings in NLP. Moreover, the spatial structure formed by pixels is analogous to the compositional structure of text formed by words (Fig. 1). This inherent similarity motivates us to adopt the hierarchical paradigm that has been proven effective in NLP [34]. In NLP, jointly modeling word- and sentence-level representations not only enables the learning of individual features but also facilitates the modeling of their interactions. More fundamentally, this hierarchical perspective aligns with the intrinsic multiscale structure of hyperspectral data. In the spectral domain, contiguous bands can be naturally grouped by physical absorption features or spectral intervals, forming a hierarchy from fine-grained band-level details to coarse-grained material semantics. This aligns well with hierarchical modeling, where low-level features capture local spectral patterns and high-level features aggregate them into discriminative material representations. In the spatial domain, objects and materials appear at multiple scales, requiring models to capture both local structures and global context. A hierarchical architecture is therefore well-suited to jointly represent these nested spectral–spatial patterns.

In contrast, conventional dual-branch architectures in HSIC typically process spectral and spatial information through separate encoders, with fusion occurring only at a late stage. This separation breaks the natural coupling between a pixel's spectral signature and its spatial context, as these two modalities are inherently intertwined in hyperspectral data. Specifically,

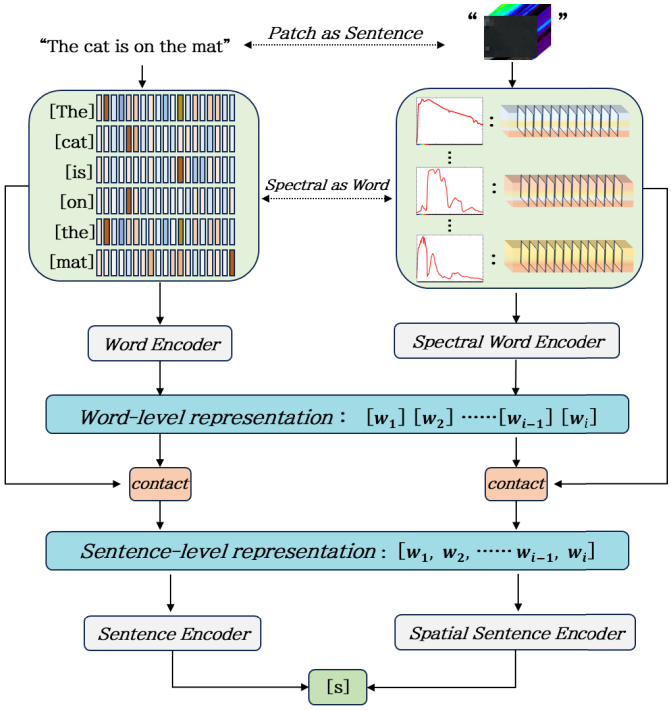


Fig. 1. Analogy between the compositional structure of spectral–spatial features in hyperspectral images and that of sentences formed by words.

the spectral reflectance of a pixel is physically determined not only by its material properties but also by its surrounding environment (e.g., mixed pixels, illumination variations, and spatial adjacency). By treating spectral and spatial features as independent pathways, dual-branch models may lose the opportunity to capture such cross-modal interactions during early feature extraction, thereby disrupting the intrinsic physical correlations.

To address this limitation, we propose a unified hierarchical Mamba-based HSIC network that jointly models spectral–spatial features from the outset, thereby preserving their natural coupling. The proposed network, named language-inspired hierarchical mamba network with superpixel-structured training (L-HMambaS), employs a hierarchical Mamba architecture to jointly model spectral–spatial relationships. First, we introduce a regional clustering-based superpixel sampling (RCSS) method to generate more representative and homogeneous training patches under limited labeled samples. This ensures that each patch centers on a semantically coherent region, preserving object boundaries, reducing mixed-pixel effects, and enhancing spatial consistency. Second, we design a learnable scanning module that fundamentally differs from conventional multidirectional scanning strategies, which rely on fixed patterns and require repeated scanning of each sample. Instead, our module leverages the representative center pixel obtained from RCSS to compute similarity scores with other pixels in the patch and then arranges the pixels into a sequence ordered from high to low similarity. This adaptive scanning strategy preserves discriminative spectral–spatial relationships while significantly reducing training time. Finally, as illustrated in Fig. 1, inspired by

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hierarchical feature modeling in NLP [34], we note that pixel sequences formed through scanning can be viewed as spatial arrangements of spectral vectors and define the compositional structure of spectral–spatial features as follows. Each pixel's spectral signature serves as elementary, location-independent atomic unit, while spatial arrangements define a set of composition rules (such as adjacency, spectral similarity-based grouping, and hierarchical aggregation) that systematically combine these atomic units into larger, semantically meaningful entities (e.g., homogeneous regions, object parts, or complete objects). This compositional view is inherent to hyperspectral data because pixels of the same material tend to form spatially contiguous areas with highly similar spectral profiles, while material boundaries induce abrupt changes. In analogy to natural language, where words (primitives) combine via syntactic rules to form phrases and sentences (composed structures), our method treats spectral vectors as "words" and spatial relations as "composition rules" to jointly model spectral–spatial hierarchies. Drawing further from hierarchical contextualized representation [34], our model introduces a pure Mamba-based hierarchical encoder built upon inner/outer Mamba blocks. This enables direct and integrated modeling of spectral–spatial features within a unified architecture.

To summarize, the main contributions of our work are as follows.

1) *RCSS:* A superpixel-based sampling strategy designed to mitigate sample space overlap and obtain semantically homogeneous image patches.
2) *Semantic Cross-Scan (SCS):* A learnable scanning mechanism that arranges semantically similar pixels contiguously within sequences, thereby eliminating the need for repeated training of the same sample under multiple fixed scanning directions.
3) *Pure Mamba-Based Hierarchical Architecture:* Departing from conventional dual-branch architectures, we propose a hierarchical Mamba block composed of inner/outer Mamba units to directly model fused spectral–spatial features. Additionally, we explore the relationship between spectral and spatial domains from a linguistic perspective.

In Section IV, we conduct extensive experiments on five public hyperspectral image datasets. The experimental results show that our proposed method achieves competitive performance, surpassing all competing models.

The remainder of this article is organized as follows. Section II introduces the related work on HSIC and superpixel segmentation issues. Section III presents the architecture of the proposed L-HMambaS model in a step-by-step fashion. Section IV presents a comprehensive experimental assessment of the L-HMambaS model. Section V concludes this article with some remarks and hints at plausible future research lines.

II. Related Work

A. Hyperspectral Image Classification

In recent years, HSIC research has predominantly concentrated on CNN-based [8], [10], [11], [35], [36], [37], Transformer-based [15], [16], [17], [18], hybrid CNN–Transformer frameworks [20], [21], and, more recently, Mamba-based approaches.

1) CNN-Based Models for HSIC: CNN-based approaches have demonstrated strong capabilities in extracting local spectral–spatial features. For instance, DCNN [8] leverages local spatial–spectral relationships among neighboring pixels to effectively incorporate contextual information. Similarly, the framework in [10] employs a series of convolutional and pooling layers to extract nonlinear, discriminative, and invariant deep features from hyperspectral images. CLOLN [11] proposes a dual-single-channel 3-D convolution coupled with depthwise convolution to comprehensively extract spectral–spatial information. ACL-CNN [35] utilizes adversarial complementary learning to disentangle common and specific patterns across different modalities, while S2PNet [36] separately captures spatial and spectral features via CNNs and integrates them through a memory unit. More recently, SpectralX [37], [38] presents a parameter-efficient fine-tuning framework that adapts optically pretrained remote sensing foundation models to multispectral/hyperspectral data using a Hyper Tokenizer and attribute-oriented adapters, achieving notable domain generalization performance.

2) Transformer-Based Models for HSIC: Transformer-based architectures achieve global spectral–spatial modeling through self-attention mechanisms. For example, SpectralFormer [15] learns spectrally local sequence information from neighboring bands in hyperspectral images, yielding groupwise spectral embeddings. SSFTT [16] introduces a spectral–spatial feature tokenization transformer that captures both low-level spectral–spatial features and high-level semantic representations. GAHT [17] proposes a group-aware hierarchical Transformer, which confines multihead self-attention to local spatial–spectral contexts via a grouped pixel embedding module. GSC-ViT [18] incorporates a group-wise separable convolution (GSC) module, combining grouped pointwise convolution (GPWC) and group convolution to enhance feature extraction.

3) Hybrid CNN–Transformer Frameworks for HSIC: Synergistic architectures that combine CNNs and Transformers have been developed to leverage complementary local and global feature modeling capabilities for HSIC. For instance, FTSCN [20] integrates Transformers with SimAM-enhanced CNNs to capture intricate spatial characteristics of HSIs. CAF-Former [21] proposes a cross-attention fusion network that bridges CNNs and Transformers, effectively exploiting the local feature representation strengths of CNNs and the long-range dependency modeling of Transformers. However, existing HSIC methods still face limitations in capturing long-range dependencies due to inherent constraints, such as the local receptive fields of CNNs and the quadratic computational complexity of Transformers.

4) SSMs for HSIC: SSMs [39], originally stemming from control theory, have recently shown significant promise in deep learning for sequence modeling [40]. Works such as [26], [27], and [28] have applied Mamba to HSIC. MambaHSI [26] stacks cascaded Mamba blocks with parallel spectral and spatial branches to extract complementary features. IGroupSS [27] employs an IGSM mechanism that splits the spectral dimension into four regular groups and applies unidirectional

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S6 operators per group to reduce spectral redundancy before global modeling. HSIMFormer [28] further hybridizes Transformer and Mamba via an ITE-CME design to leverage their complementary strengths.

While both Transformer and Mamba were initially developed for natural language processing, they have recently achieved notable success in HSIC. Moreover, hierarchical feature modeling has proven highly effective in visual tasks [34]. In this context, we propose a novel hierarchical Mamba architecture for HSIC, distinct from conventional dual-branch structures.

B. Regional Clustering-Based SLIC

Superpixel segmentation is a standard preprocessing step for image abstraction [41], and it excels at structuring images while retaining object boundaries. Recent deep learning pipelines have systematically exploited this property [42], [43], [44]. In these endeavors, superpixels are employed to provide more semantically consistent region partitioning instead of fixed grid inputs, thereby suppressing within-region redundancy and injecting mid-level semantic structure into deep learning architectures [41], [45].

The regional clustering-enhanced simple linear iterative clustering (RC-SLIC) algorithm [46] extends SLIC [47] to hyperspectral superpixel segmentation. While SLIC operates on a 5-D vector 3-D CIELAB color + 2-D spatial coordinates), RC-SLIC replaces the color term with a spectral metric. Specifically, the SID-SAM [48] distance is adopted as it simultaneously quantifies spectral variability and pixel-wise similarity. Let $x_i = (x_{i1}, x_{i2}, \ldots, x_{il})$ denote a given pixel with spatial coordinate (i_x, i_y) in the HSI, and its corresponding cluster center of $x_c = (x_{c1}, x_{c2}, \ldots, x_{cl})$ is located at spatial coordinate (c_x, c_y). The clustering criterion $D_{i,c}$ is defined by the following equation:

$$\begin{aligned} D_{i,c} &= (1-\lambda) \times D_{spectral} + \lambda \times D_{spatial} \\ D_{spectral} &= SID(x_i, x_c) \times \tan(SAM(x_i, x_c)) \\ D_{spatial} &= \sqrt{(i_x - c_x)^2 + (i_y - c_y)^2} \Big/ r \end{aligned} \tag{1}$$

where $D_{spectral}$ denotes the spectral distance and $SID(x_i, x_c)$ and $SAM(x_i, x_c)$ refer to the spectral information divergence (SID) and spectral angle mapper [48]. $D_{spatial}$ denotes the spatial Euclidean distance (ED) with normalization, r is the length of the diagonal for the search neighborhood, and λ is a weighting factor used to balance the spectral and spatial terms and the balance parameter λ is suggested to be less than 0.5.

III. Proposed Methodology

This section presents a detailed explanation of our proposed L-HMambaS model. We first elaborate on the overall architecture of the model and its design motivation. Subsequently, we delve into the implementation details of some key components within the model.

A. Overall Architecture

As observed from Fig. 2, the input HSI is first oversegmented into superpixels by RC-SLIC. Compared with conventional SLIC, RC-SLIC supports multiple hyperspectral-specific metrics, including spectral angle distance (SAD) and SID. These metrics effectively capture the shape characteristics of spectral signatures and are beneficial for preserving spectral homogeneity. Each superpixel center provided by RCSS crops a $P \times P \times C$ patch whose pixels are spectrally coherent; this guarantees that the central pixel is embedded in a homogeneous spatial context.

Subsequently, these patches are fed into the convolutional dimensionality reduction block (ConvDR) block, which comprises Conv1D and Conv2D. Conv1D slides along the spectral dimension, capturing local correlations among spectral bands and performing nonlinear feature extraction, as well as dimensionality reduction. Concurrently, Conv2D acts on the spatial dimensions, aggregating spatial context.

After the spectral–spatial rearrangement by the SCS operation, the resulting sequence–interpreted as a spatial sentence composed of spectral words–enters a customized hierarchical Mamba block. In this block, the residual-equipped Mamba encoder is employed to capture long-range dependencies across the sequence. Since hyperspectral images inherently possess a natural hierarchical structure—just as text evolves from characters to words, sentences, and paragraphs—hyperspectral data progress from individual spectral bands to single-pixel spectra and ultimately to patches composed of multiple pixels, the hierarchical Mamba block serves as a pure Mamba-based hierarchical encoder. Concretely, the pixel is fed into an inner spectral Mamba [inner word-level mamba (ISM)] to generate word-level embedding representations. These word-level representations are then used to enhance pixel sequence by injecting into an outer spatial Mamba (OSM) that yields sentence-level embedding representations. Finally, the sentence-level representations are fed into a multilayer perceptron (MLP) for classification purposes.

B. Regional Clustering-Based Superpixel Sampling

Unlike conventional grid or random patch sampling strategies, we propose a superpixel-based sampling method (termed RCSS) that yields patches with high intraregion homogeneity and minimal overlap, cutting redundancy and enhancing feature fidelity.

As illustrated in Fig. 3, given the hyperspectral image $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$, where H and W denote the height and width and C represents the number of spectral bands, after one iteration of the RC-SLIC, the image $\mathbf{I}$ is partitioned into K superpixels, and we index them by $k \in \{0, 1, \ldots, K-1\}$. The pixel labels are represented as labels $\in \mathbb{Z}^{H \cdot W}$, indicating the superpixel assignment for each pixel. The set of pixels belonging to superpixel k is defined as follows:

$$\text{pts}_k = \{i \mid \text{labels}[i] = k\}. \tag{2}$$

Let $\mathbf{p}_i$ denote the spectral vector of pixel i and $\mathbf{s}_k$ represent the mean spectrum of superpixel k. For pixel $\mathbf{p}_i$ with spatial coordinates (x_i, y_i), its similarity weight with respect to $\mathbf{s}_k$ is computed by the following equation:

$$w_i = \frac{1}{d(\mathbf{p}_i, \mathbf{s}_k) + \epsilon}, \quad \epsilon = 10^{-6}$$

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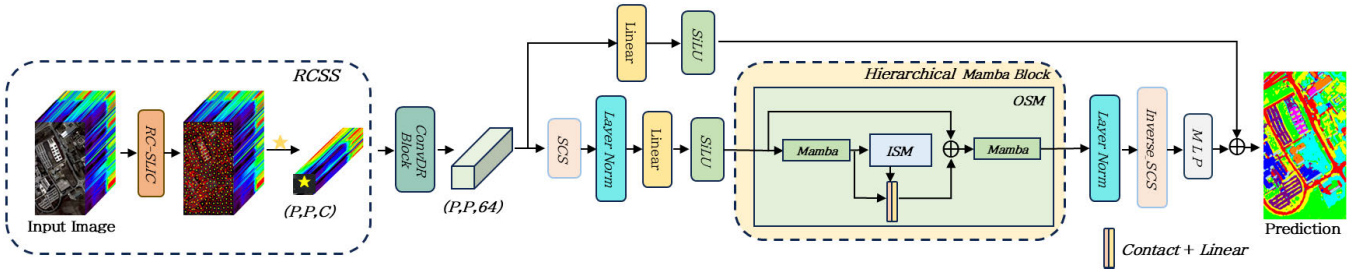


Fig. 2. Schematic of the proposed L-HMambaS network. "RC-SLIC" is the regional clustering-based SLIC module for superpixel segmentation of hyperspectral image. "RCSS" is regional clustering-based superpixel sampling. The ⋆ represents the center pixel of superpixel. "ConvDR Block" is a convolutional dimensionality reduction block. "SCS" is a semantic cross-scan module. "hierarchical Mamba block" denotes a pure Mamba-based hierarchical encoder consisting of an OSM and an ISM, performing spectral–spatial semantic feature fusion. The outputs are subsequently reordered according to the scanning type, thereby restoring their original spatial structure. "MLP" is a multilayer perceptron layer used for final classification. ("⊕") represents addition.

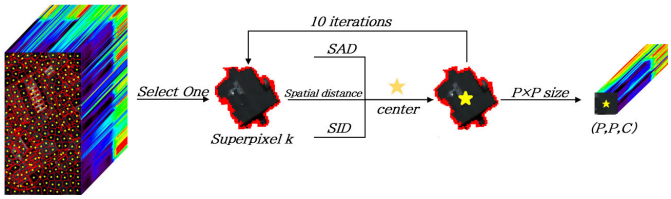


Fig. 3. Process of RCSS.

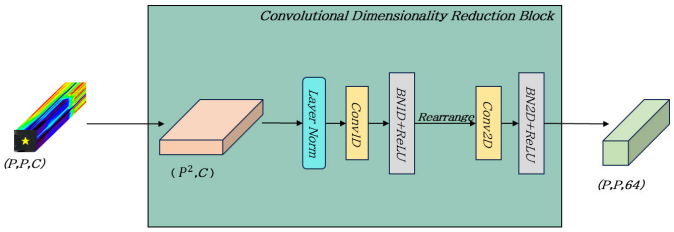


Fig. 4. Process of ConvDR.

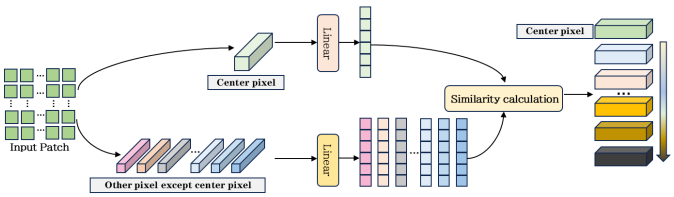


Fig. 5. Process of SCS block.

$$d(\mathbf{p}_i, \mathbf{s}_k) = \mathrm{SAD}(\mathbf{p}_i, \mathbf{s}_k) + \mathrm{SID}(\mathbf{p}_i, \mathbf{s}_k). \tag{3}$$

Then, we normalize the weights of all pixels by using the following equation:

$$w_i \leftarrow \frac{w_i}{\sum_{m \in \mathrm{pts}_k} w_m}, \quad i \in \mathrm{pts}_k \tag{4}$$

and the center $(\mathrm{seed}_x[k], \mathrm{seed}_y[k])$ of superpixel k is computed using the normalized weights as follows:

$$\mathrm{seed}_x[k] = \sum_{i \in \mathrm{pts}_k} w_i \cdot x_i, \quad \mathrm{seed}_y[k] = \sum_{i \in \mathrm{pts}_k} w_i \cdot y_i. \tag{5}$$

Finally, with each iteration of the RC-SLIC, the mean spectrum of superpixel k, denoted as $\mathbf{s}_k = \mathrm{seed_specs}[k]$, is updated by averaging the spectral vectors of all pixels within pts_k

$$\mathbf{s}_k[j] = \frac{1}{|\mathrm{pts}_k|} \sum_{i \in \mathrm{pts}_k} \mathbf{p}_{[i]}[j], \quad j = 1, 2, \ldots, C. \tag{6}$$

By doing so, the pixels within each patch exhibit high homogeneity while minimizing the overlap between different patches.

C. Convolutional Dimensionality Reduction Block

As shown in Fig. 4, the center-cropped patch $\mathbf{X}_{\mathrm{input}} \in \mathbb{R}^{P \times P \times C}$ is reshaped into $\mathbf{X}_{\mathrm{reshaped}} \in \mathbb{R}^{P^2 \times C}$ and layer-normalized for training stability. Then, $\mathbf{X}_{\mathrm{reshaped}}$ is fed into the Conv1D, and undergoes a point-wise convolution along the spectral dimension. Here, the Conv1D uses a convolution kernel $K_1 \in \mathbb{R}^{1 \times 1 \times 3}$, which focuses on capturing local information within the spectral dimension and extracting the relationships between adjacent bands of each pixel. The procedure can be formulated by the following equation:

$$\begin{aligned} \mathbf{X}_{\mathrm{L}} &= \mathrm{LayerNorm}\left(\mathbf{X}_{\mathrm{reshaped}}\right) \\ \mathbf{X}_{\mathrm{Conv1d}} &= \mathrm{ReLU}\left(\mathrm{BN1D}\left(\mathrm{Conv}\left(K_1, \mathbf{X}_{\mathrm{L}}\right)\right)\right). \end{aligned} \tag{7}$$

The output feature $\mathbf{X}_{\mathrm{Conv1d}}$ is reshaped back to the original spatial dimensions and forwarded to the Conv2D to capture spatial contextual features. In the Conv2D, the convolution kernel of $K_2 \in \mathbb{R}^{3 \times 3}$ is utilized to effectively extract fine-grained local details of the image while preserving spatial resolution. Since the kernel size is odd, the padding is set to half of the kernel size, thereby ensuring that the convolution operation maintains the spatial dimensions of the input feature map unchanged

$$\begin{aligned} \mathbf{X}_{\mathrm{mid}} &= \mathrm{Reshape}\left(\mathbf{X}_{\mathrm{Conv1d}}\right) \\ \mathbf{X}_{\mathrm{Conv2d}} &= \mathrm{ReLU}\left(\mathrm{BN2D}\left(\mathrm{Conv}\left(K_2, \mathbf{X}_{\mathrm{mid}}\right)\right)\right). \end{aligned} \tag{8}$$

Conv1D employs learnable kernels to automatically capture local spectral features, performing nonlinear dimensionality reduction and explicitly modeling interband correlations, thereby enhancing the discriminative power of spectral representations. Compared with PCA, this procedure is fully learnable and more flexible.

D. Semantic Cross-Scan

As shown in Fig. 5, the input patch is decomposed into the center pixel $\mathbf{c}$ and the remaining $P^2 - 1$ neighboring pixels set

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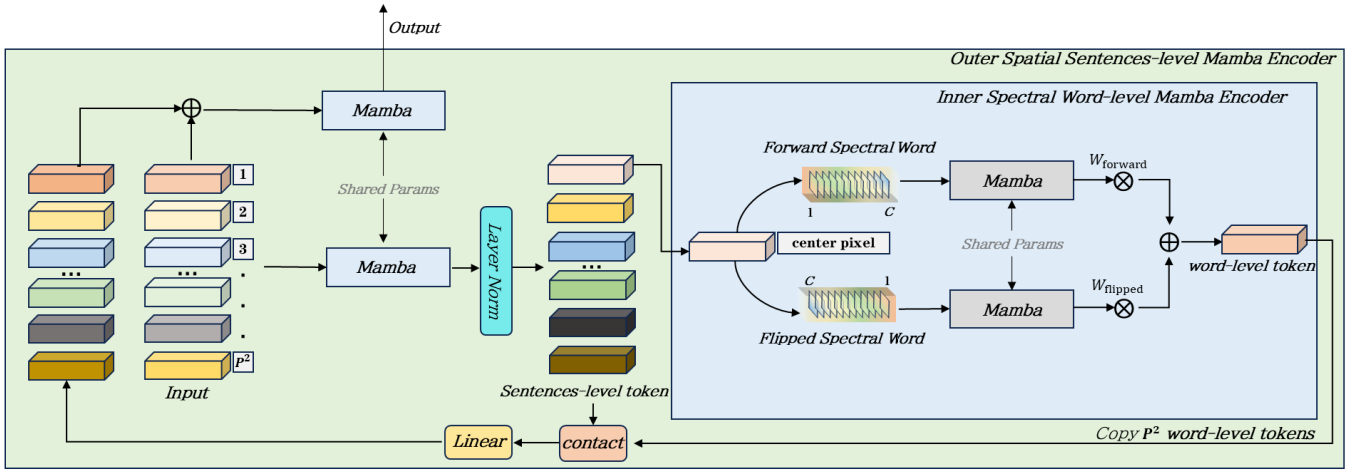


Fig. 6. Structure of hierarchical Mamba block. All ISM blocks share parameters. The spatial sentences are then fused with word-level representations, and the enhanced features are further processed by the OSM block. ("⊗") represents element-wise multiplication and ("⊕") represents addition.

O. Two learnable linear projections are applied as shown in the following equation:

$$\begin{aligned}\mathbf{q} &= W_q\mathbf{c} \in \mathbb{R}^{B\times D}\\ \mathbf{K} &= W_k\mathbf{O}^\top \in \mathbb{R}^{B\times(P^2-1)\times D}\end{aligned} \tag{9}$$

where $W_q, W_k \in \mathbb{R}^{C\times D}$ are trainable parameters and D denotes the projection dimension. The scaled dot-product similarity between the center pixel and each neighboring pixel is then computed with the following equation:

$$\mathbf{s} = \frac{\mathbf{q}\mathbf{K}^\top}{\sqrt{D}} \in \mathbb{R}^{B\times(P^2-1)}. \tag{10}$$

Formula (10) is adapted from the scaling technique used in Transformer attention mechanisms, so as to stabilize gradients in high-dimensional dot-product operations. The output sequence of pixels is ordered in descending similarity to the center pixel, which remains fixed at the first position. The original positional mapping for each element in this sequence is preserved to facilitate the subsequent InverseSCS operation.

E. Hierarchical Mamba Block

The detailed structure of the hierarchical Mamba block is depicted in Fig. 6. It comprises two nested units: the ISM block, which operates on spectral words, and the outer sentence-level Mamba (OSM) block, which processes the spatial sentences.

The input feature map $\mathbf{X} \in \mathbb{R}^{P\times P\times C}$ is first processed by the SCS module. Prior to their injection into the hierarchical Mamba block, the following stabilizing transformations are applied:

$$\mathbf{X}_{\text{input}} = \text{SiLU}\left(\text{Linear}\left(\text{LayerNorm}\left(\text{SCS}\left(\mathbf{X}\right)\right)\right)\right). \tag{11}$$

In the hierarchical Mamba block, the flattened sequence $\mathbf{X}_{\text{input}} \in \mathbb{R}^{P^2\times C}$ is first input into OSM to generate spatial feature $\mathbf{S} \in \mathbb{R}^{P^2\times C}$

$$\mathbf{S} = \text{LayerNorm}\left(\text{Mamba}\left(\mathbf{X}_{\text{input}}\right)\right). \tag{12}$$

Exploiting the fact that each pixel of a hyperspectral image encompasses tens to hundreds of spectral bands, we naturally treat the spectrum of a single pixel as a word $\mathbf{W} \in \mathbb{R}^C$, the spatial feature is then ordered as follows:

$$\mathbf{S} = \left[\mathbf{W}^1, \mathbf{W}^2, \ldots, \mathbf{W}^{P^2}\right] \tag{13}$$

where $\mathbf{W}^i$ denotes the ith pixel in sequence $\mathbf{S}$. Owing to the RCSS strategy, most pixels within the outer stream $\mathbf{S}$ are spectrally homogeneous, making the center pixel highly representative of the sequence $\mathbf{S}$. Therefore, only the center pixel is passed into the ISM as the inner stream. While a single spectral vector does not constitute a long sequence, allowing the inner module to be implemented by a 1D-CNN or a Mamba-based architecture, we adopt Mamba to fully exploit the near-continuous sequential nature of hyperspectral data along the spectral dimension. This design enables effective modeling of long-range spectral dependencies, as verified in the ablation studies in Section IV-E1. Consequently, even with minimal input, the SSM-equipped block captures rich spectral context and outputs contextually enriched tokens.

To enhance spectral sequence modeling, ISM adopts a bidirectional spectral Mamba structure by processing the spectral sequence in both the forward and flipped orders. The forward central pixel is defined as follows:

$$\overrightarrow{\mathbf{W}}^{ctr} = \left[w_1, w_2, \ldots, w_c\right] \tag{14}$$

and the flipped central pixel is defined as follows:

$$\overleftarrow{\mathbf{W}}^{ctr} = \left[w_c, w_{c-1}, \ldots, w_1\right]. \tag{15}$$

The bidirectional sequences $\overrightarrow{\mathbf{W}}^{ctr}$ and $\overleftarrow{\mathbf{W}}^{ctr}$ are subsequently fed into the Mamba module, whose inner SSM dynamically captures dependencies between bands along spectral dimensions. Through leveraging its dynamic nature, the SSM enhances the model's ability to perceive spectral information and improves its capacity to distinguish between different land-cover categories. Subsequently, the word-level token $W_t^{ctr} \in \mathbb{R}^C$ is obtained as a weighted combination of $\overrightarrow{\mathbf{W}}^{ctr}$ and $\overleftarrow{\mathbf{W}}^{ctr}$. Since the two directions capture complementary spectral

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dependency patterns, directly assigning them equal importance may limit the model flexibility. Therefore, we introduce two learnable fusion coefficients, ω_{fwd} and $\omega_{flipped}$, to adaptively balance the contributions of the two branches. In this way, the network can automatically adjust the importance of forward and flipped spectral representations according to the training data, leading to more effective bidirectional spectral feature aggregation. The fusion process can be formulated by the following equation:

$$\mathbf{W}_t^{ctr} = \omega_{fwd} \times \text{Mamba}\left(\overrightarrow{\mathbf{W}}^{ctr}\right) + \omega_{flipped} \times \text{Mamba}\left(\overleftarrow{\mathbf{W}}^{ctr}\right). \tag{16}$$

At the sentence level, the outer stream $\mathbf{S}$ is augmented by the word-level tokens. Concretely, word-level token $\mathbf{W}_t^{ctr}$ is concatenated with every original pixel $\mathbf{W}$ and then combined through a linear projection. Since $\mathbf{W}_t^{ctr}$ is highly representative, this design allows it to function as a form of long-term memory, influencing every time step in the OSM modeling process. We view the enhanced feature sequence as $\mathbf{S}_{\text{enh}} \in \mathbb{R}^{P^2 \times C}$, as formulated in the following equation:

$$\mathbf{W}_{\text{enh}}^i = \text{Linear}\left(\left[\mathbf{W}_t^{ctr}; \mathbf{W}^i\right]\right)$$
$$\mathbf{S}_{\text{enh}} = \left[\mathbf{W}_{\text{enh}}^1, \mathbf{W}_{\text{enh}}^2, \ldots, \mathbf{W}_{\text{enh}}^{P^2}\right]. \tag{17}$$

The enhanced representation is summed with the original input via a residual connection to obtain $\mathbf{S}_{re} \in \mathbb{R}^{P^2 \times C}$. We use OSM to transform the enhanced sentence embedding again. The OSM can model the pairwise interactions between word embeddings. Its output is a compact sentence-level token $\mathbf{S}_t \in \mathbb{R}^{P^2 \times C}$ that encodes global spectral–spatial context

$$\mathbf{S}_{re} = \mathbf{S}_{enh} + \mathbf{X}_{input}$$
$$\mathbf{S}_t = \text{LayerNorm}\left(\text{Mamba}\left(\mathbf{S}_{re}\right)\right). \tag{18}$$

Prior to MLP, the sentence-level token is reconstructed into its original 2-D spatial layout of $\mathbf{Y}_t \in \mathbb{R}^{P \times P \times C}$

$$\mathbf{Y}_t = \text{InverseSCS}\left(\mathbf{S}_t\right). \tag{19}$$

The fused representation is finally summed with the original input via a residual connection [see (20)] to obtain the input of MLP $\mathbf{X}_{re} \in \mathbb{R}^{P \times P \times C}$, systematically preserving low-level details while enabling full cross-dimensional feature interaction. This deep fusion mechanism amplifies the model's capability to exploit complex dependencies and consistently boosts classification accuracy

$$\mathbf{X}_{re} = \mathbf{Y}_t + \text{SiLU}\left(\text{Linear}\left(\mathbf{X}_{\text{Conv2d}}\right)\right). \tag{20}$$

All in all, the hierarchical Mamba block strengthens long-range dependency modeling by jointly reasoning over spectral and spatial sequences while explicitly preserving their intrinsic coupling, leading to consistent gains in the accuracy and robustness of HSIC.

IV. Experiments

In this section, we first describe the experimental setup, including datasets, implementation details, and evaluation metrics. Then, we conduct a detailed ablation study to analyze the effect of the proposed modules. Finally, we comprehensively compare with state-of-the-art methods both qualitatively and quantitatively.

TABLE I
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CLASS IN THE PAVIAU DATASET

No.	Color	Class Name	Train	Valid	Test
C1		Asphalt	30	20	6581
C2		Meadows	30	20	18599
C3		Gravel	30	20	2049
C4		Trees	30	20	3014
C5		Painted metal sheets	30	20	1295
C6		Bare Soil	30	20	4979
C7		Bitumen	30	20	1280
C8		Self-Blocking Bricks	30	20	3632
C9		Shadows	30	20	897
		Total	270	180	42326

TABLE II
NUMBER OF TRAINING, VALIDATION, AND TEST SAMPLES FOR EACH CLASS IN THE WHU-HI-LONGKOU DATASET

No.	Color	Class Name	Train	Valid	Test
C1		Corn	30	20	34461
C2		Cotton	30	20	8324
C3		Sesame	30	20	2981
C4		Broad-leaf soybean	30	20	63162
C5		Narrow-leaf soybean	30	20	4101
C6		Rice	30	20	11804
C7		Water	30	20	67006
C8		Roads and houses	30	20	7074
C9		Mixed weed	30	20	5179
		Total	270	180	204092

A. Hyperspectral Datasets

To validate the effectiveness of L-HMambaS, we adopt five publicly hyperspectral datasets for comprehensive evaluation: Pavia University [49], WHU-Hi-LongKou, WHU-Hi-HanChuan, WHU-Hi-HongHu [50], [51], and Houston 2013 [52]. Detailed descriptions are provided as follows.

1) Pavia University: The Pavia University dataset, abbreviated as PaviaU, was acquired by the ROSIS sensor over Pavia, northern Italy. This dataset contains 103 spectral bands and spatial dimensions of 610 × 340 pixels, with a spatial resolution of 1.3 m. It includes nine land-cover classes, such as asphalt, grass, gravel, etc. The division of training, verification, and testing sets is shown in Table I, with approximately 30 training samples and 20 verification samples per class.

2) WHU-Hi-LongKou: The dataset, abbreviated as WHU-LK, was collected in LongKou, Hubei, China, using a Headwall Nano-Hyperspec imaging sensor with an 8-mm focal length, mounted on a DJI Matrice 600 Pro UAV platform. The study area represents a simple agricultural scene, including water bodies, roads, and six types of crops. The image has a size of 550 × 400 pixels, consisting of 270 spectral bands ranging from 400 to 1000 nm, with a spatial resolution of approximately 0.463 m. The division of training, verification, and testing sets is shown in Table II, with each class containing 30 training samples and 20 verification samples per class.

3) WHU-Hi-HanChuan: The dataset, abbreviated as WHU-HC, was collected in HanChuan, Hubei, China, using a

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TABLE III

NUMBER OF TRAINING, VALIDATION, AND TEST SAMPLES FOR EACH CLASS IN THE WHU-HI-HANCHUAN DATASET

No.	Color	Class Name	Train	Valid	Test
C1		Strawberry	30	20	44685
C2		Cowpea	30	20	22703
C3		Soybean	30	20	10237
C4		Sorghum	30	20	5303
C5		Water spinach	30	20	1150
C6		Watermelon	30	20	4483
C7		Greens	30	20	5853
C8		Trees	30	20	17928
C9		Grass	30	20	9419
C10		Red roof	30	20	10466
C11		Gray roof	30	20	16861
C12		Plastic	30	20	3629
C13		Bare soil	30	20	9066
C14		Road	30	20	18510
C15		Bright object	30	20	1086
C16		Water	30	20	75351
		Total	270	180	256730

TABLE IV

NUMBER OF TRAINING, VALIDATION, AND TEST SAMPLES FOR EACH CLASS IN THE WHU-HI-HONGHU DATASET

No.	Color	Class Name	Train	Valid	Test
C1		Red roof	30	20	13991
C2		Road	30	20	3462
C3		Bare soil	30	20	21771
C4		Cotton	30	20	163235
C5		Cotton firewood	30	20	6168
C6		Rape	30	20	44507
C7		Chinese cabbage	30	20	24053
C8		Pakchoi	30	20	4004
C9		Cabbage	30	20	10769
C10		Tuber mustard	30	20	12344
C11		Brassica parachinensis	30	20	10965
C12		Brassica chinensis	30	20	8904
C13		Small Brassica chinensis	30	20	22457
C14		Lactuca sativa	30	20	7306
C15		Celtuce	30	20	952
C16		Film covered lettuce	30	20	7212
C17		Romaine lettuce	30	20	2960
C18		Carrot	30	20	3167
C19		White radish	30	20	8662
C20		Garlic sprout	30	20	3436
C21		Broad bean	30	20	1278
C22		Tree	30	20	3990
		Total	660	440	385593

Headwall Nano-Hyperspec imaging sensor with a 17-mm focal length, mounted on a Leica Aibot X6 UAV V1 platform. The study area represents an urban–rural transition zone, including buildings, water bodies, farmland, and seven types of crops. The image has a size of 1217 × 303 pixels, consisting of 274 spectral bands ranging from 400 to 1000 nm, with a spatial resolution of approximately 0.109 m. The division of training, verification, and testing sets is shown in Table III, with each class containing 30 training samples and 20 verification samples per class.

4) WHU-Hi-HongHu: The dataset, abbreviated as WHU-HH, was collected in HongHu, Hubei, China, using a Headwall Nano-Hyperspec imaging sensor with a 17-mm focal length, mounted on a DJI Matrice 600 Pro UAV platform. The study area represents a complex agricultural scene, containing multiple crops and different varieties of the same crop. The image has a size of 940 × 475 pixels, consisting of 270 spectral bands ranging from 400 to 1000 nm, with a spatial resolution of approximately 0.043 m. The division of training, verification, and testing sets is shown in Table IV, with each class containing 30 training samples and 20 verification samples per class.

TABLE V

NUMBER OF TRAINING, VALIDATION, AND TEST SAMPLES FOR EACH CLASS IN THE HOUSTON 2013 DATASET

No.	Color	Class Name	Train	Valid	Test
C1		Healthy grass	30	20	1201
C2		Stressed grass	30	20	1204
C3		Synthetic grass	30	20	647
C4		Trees	30	20	1194
C5		Soil	30	20	1192
C6		Water	30	20	275
C7		Residential	30	20	1218
C8		Commercial	30	20	1194
C9		Road	30	20	1202
C10		Highway	30	20	1177
C11		Railway	30	20	1185
C12		Parking Lot 1	30	20	1183
C13		Parking Lot 2	30	20	419
C14		Tennis Court	30	20	378
C15		Running Track	30	20	610
		Total	450	300	14279

5) Houston 2013: The dataset, abbreviated as Hs 2013, was collected using an airborne hyperspectral imaging sensor over Houston, TX, USA. This image has a size of 349× 1905 pixels, containing of 144 spectral bands ranging from 380 to 1050 nm, and with a spatial resolution of 2.5 m. It provides a rigorous testing platform for HSIC in complex urban environments. The dataset includes 15 land-cover classes, such as healthy grass, buildings, roads, railways, etc. The division of training, verification, and testing sets is shown in Table V.

B. Implementation Details

We introduce the parameters setting used in the experiments. Before model training, we apply data augmentation techniques on the input image patches, including horizontal or vertical flips, spectral noise, spectral scale, and random rotations of 90°, 180°, or 270°. Each original training sample can generate up to six-fold augmented data, where 30 training samples per class are expanded to 180 samples per class through data augmentation. We use the Adam optimizer for parameter updates, with a total of 120 training epochs. The learning rate is set to 0.0002, the weight decay coefficient is 0.0005, and the batch size is 256. The loss function used in the model is cross-entropy loss.

We use overall accuracy (OA), average accuracy (AA), and κ coefficient to evaluate classification performance. To mitigate the impact of randomness and ensure the reliability of the results, we conduct each experiment ten times with different random seeds and record the average result. The larger the mean value and the smaller the standard deviation of these metrics, the better the performance of the model. It is worth noting that all our experiments are conducted on CentOS Linux release 7.9.2009 equipped with a NVIDIA A40 GPU and 48 GB of memory, using Python 3.10 and PyTorch 2.1.1.

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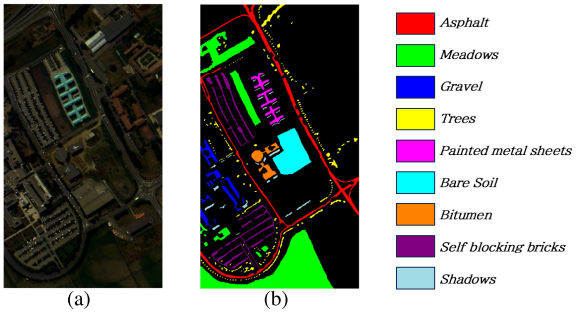


Fig. 7. (a) False-color composition of the Pavia University scene. (b) Ground-truth map containing nine mutually exclusive land-cover classes.

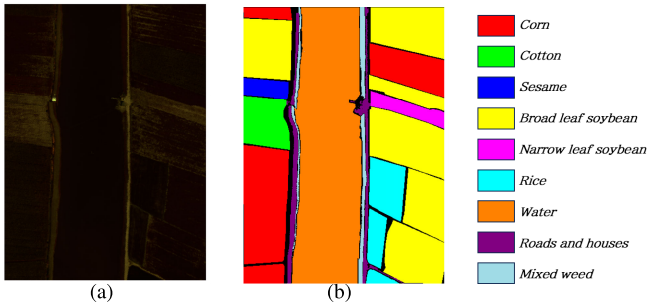


Fig. 8. (a) False-color composition of the WHU-Hi-LongKou scene. (b) Ground-truth map containing nine mutually exclusive land-cover classes.



Fig. 9. (a) False-color composition of the WHU-Hi-HanChuan scene. (b) Ground-truth map containing 16 mutually exclusive land-cover classes.

C. Analysis on Different Numbers of Superpixel Regions and Ablation Study on the RCSS

In this experiment, it is essential to select an appropriate number of partitions in accordance with the spatial resolution of hyperspectral images. The segmentation process can be interpreted as mapping spatial structural information onto the irregular boundaries of superpixel regions. Increasing the number of superpixels reduces the area of each region, thereby decreasing the spatial structural information contained within each region. Conversely, larger superpixel regions compromise intrinsic homogeneity.

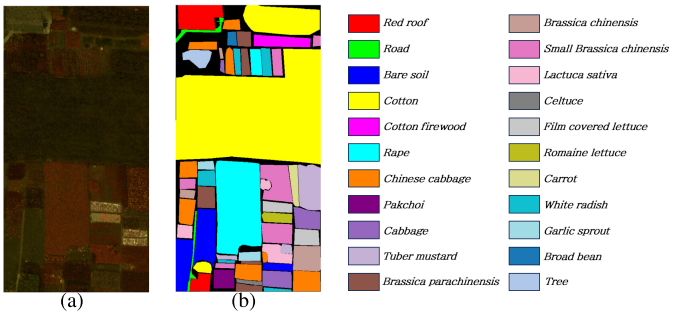


Fig. 10. (a) False-color composition of the WHU-Hi-HongHu scene. (b) Ground-truth map containing 22 mutually exclusive land-cover classes.

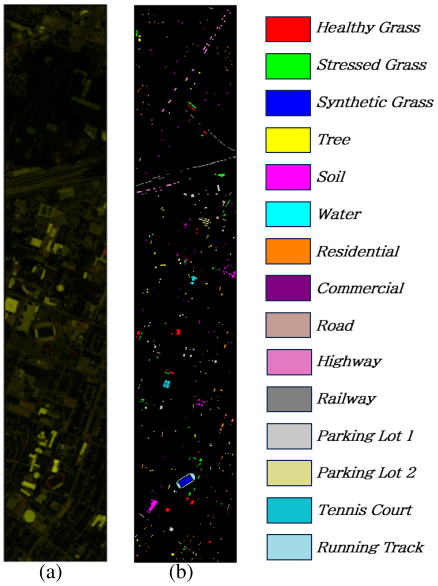


Fig. 11. (a) False-color composition of the Houston 2013 scene. (b) Ground-truth map containing 15 mutually exclusive land-cover classes.

In our experiment, we evaluated OA for different numbers of superpixel regions across five datasets with patch size 11×11. As shown in Fig. 12, the visualization results clearly demonstrate that the optimal performance on different datasets requires different numbers of segmented superpixels, which is consistent with the variations in spatial resolution among the datasets. Here, "None" denotes that the RCSS is not applied. The results indicate that using RC-SLIC with an appropriate number of superpixel segments consistently outperforms the setting without RCSS. For instance, in the PaviaU dataset, the OA rises from 94.87% (None) to 97.01% (Number:400). However, as the number of superpixel segments continues to increase, the OA begins to decline. This degradation occurs because excessive segmentation leads to overpartitioning, preventing the algorithm from obtaining more representative pixel samples.

In the WHU-series datasets, the labeled pixels are distributed in a highly compact manner, and the spatial resolution of these datasets is significantly higher than that of the PaviaU dataset. Consequently, a much larger number of superpixel segments is required compared to PaviaU to achieve optimal performance. For instance, in WHU-LK, the OA rises from 96.75% (None) to 98.45% (Number: 2500); in WHU-HC, the

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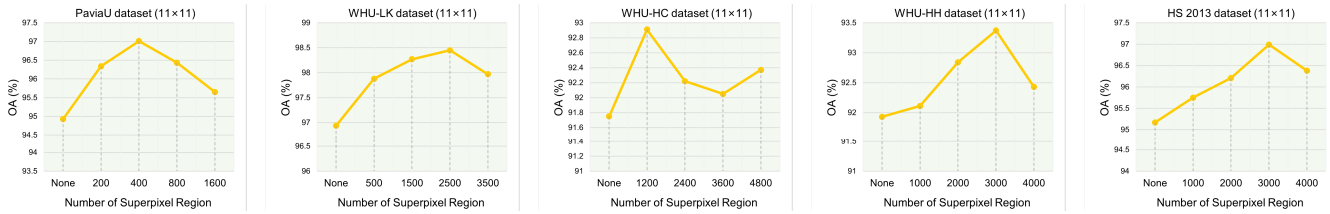


Fig. 12. OA by using different numbers of superpixel regions on five datasets.

TABLE VI
OA WITH AND WITHOUT SCS ACROSS DATASETS

Scanning Strategy	Overall Accuracy (OA)					Train time on PaviaU/s	Test time on PaviaU/s
	PaviaU	WHU-LK	WHU-HC	WHU-HH	Hs 2013		
Flatten	96.10±1.94	97.98±0.83	92.20±0.89	90.84±1.08	95.59±1.17	80.66	14.97
Multi-directional scanning	**97.11±1.55**	98.42±0.71	92.70±1.19	93.28±1.16	96.77±1.08	362.78	33.16
SCS	97.01±1.76	**98.45±0.30**	**92.91±0.28**	**93.37±0.62**	**96.79±0.57**	82.06	15.84

OA rises from 91.78% (None) to 92.91% (Number: 1200); and in WHU-HH, the OA rises from 91.89% (None) to 93.37% (Number: 3000). Notably, in the HS 2013 dataset, the labeled samples are extremely sparse, while the RCSLIC algorithm performs sampling across the entire image. When the number of superpixel segments is set too low, the probability of obtaining effective and representative samples during sampling decreases significantly. We can observe that the OA rises from 95.24% (None) to 96.79% (Number: 3000). Our method still achieves stable and superior performance compared to the baseline, demonstrating its robustness to variations in sampling granularity. Similarly, for the WHU-LK, WHU-HC, and WHU-HH datasets, when an appropriate number of superpixel blocks is adopted, our method achieves an improvement of approximately from + 1.13% to + 2.14% over the baseline.

These results clearly demonstrate that RCSS is proven effective, indicating that patches with high homogeneity are beneficial for training.

D. Ablation Study on the SCS

To validate the contribution of the proposed SCS module, we conduct an ablation study by replacing it from the full model. The quantitative results are reported in Table VI. Compared to the Flatten baseline, the proposed SCS module consistently improves OA across all five hyperspectral datasets, with gains ranging from + 0.47% of WHU-LK to + 2.53% of WHU-HH. When compared to the multidirectional scanning strategy, SCS achieves superior OA on four datasets, while being only marginally lower on the PaviaU dataset. Importantly, SCS delivers these competitive or better accuracy levels with dramatically higher computational efficiency. On the PaviaU dataset, SCS adds only minimal overhead to both training time (80.66–82.06 s) and testing time (14.97–15.84 s) relative to the Flatten baseline. In contrast, multidirectional scanning increases training time by more than 4.5× (to 362.78 s) and testing time by over 2× (to 33.16 s).

To clarify whether SCS preserves local spatial geometry and texture information, we provide both visual analysis and design justification. As shown in Fig. 13, two representative

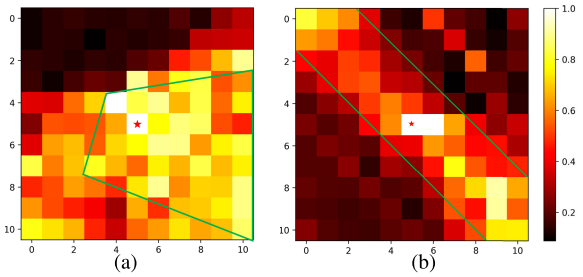


Fig. 13. Heat map of similarity between the central pixel and other pixels in different patches. The color bar represents the normalized spectral similarity values ranging from 0 to 1, where higher values indicate stronger spectral similarity. The axis ticks from 0 to 10 denote the spatial index along the side length of the patch.

patches are selected. In patch Fig. 13(a), which is largely homogeneous, the spectral similarity between the center pixel ⋆ and its neighbors is high (>0.7 for >80% of pixels), and pixels closer to the center exhibit higher similarity, reflecting natural spatial continuity. In Fig. 13(b), which contains clear structural boundaries (e.g., edges or diagonal patterns), the similarity heat map shows that high-similarity pixels are concentrated along the main diagonal, indicating a directional texture.

Importantly, SCS reorders pixels by descending similarity but does not discard original spatial coordinates. The position index of each pixel is implicitly encoded through SSM and is added to the sequence embedding before being fed into the OSM. Thus, even after reordering, the model retains full access to the original geometric layout. This design ensures that spatial context (such as edges and textures) is not lost; rather, the sequence order emphasizes spectral coherence, while positional encodings preserve geometry.

Empirically, if SCS destroyed critical spatial features, classification accuracy would degrade, especially on datasets rich in texture and fine boundaries (e.g., WHU-HC). Instead, SCS achieves superior or comparable OA to multidirectional scanning while being dramatically faster, as shown in Table VI. This confirms that SCS effectively balances spectral grouping and spatial preservation.

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TABLE VII
OA WITH AND WITHOUT ISM ACROSS DATASETS

	Overall Accuracy (OA)				
	PaviaU	WHU-LK	WHU-HC	WHU-HH	Hs 2013
Only OSM	95.79±1.95	97.87±0.83	91.08±1.04	91.93±1.52	95.77±1.55
ISM/OSM	**97.01±1.76**	**98.45±0.30**	**92.91±0.98**	**93.37±1.12**	**96.79±0.57**
1D-CNN/OSM	96.38±1.92	98.18±0.66	91.51±1.03	92.50±1.59	95.74±0.81

These results clearly demonstrate that, compared to directly flattening features, SCS significantly boosts both classification accuracy and model stability with negligible extra cost. Furthermore, when contrasted with the multidirectional scanning approach, SCS provides superior performance across most datasets while offering a much better accuracy–efficiency tradeoff. This validates SCS as a lightweight, effective, and highly efficient module for scanning strategy in HSIC.

E. Ablation Study and Visualization on Hierarchical Mamba Block

1) Ablation Study on the Branches of the ISM/OSM Blocks: To clarify the advantage of the ISM over conventional 1D-CNNs, especially given that ISM processes only the center pixel's spectral vector (a short sequence of length equal to the number of bands), we conducted an ablation study. As shown in Table VII, removing ISM and solely using OSM leads to lower accuracy and higher variance across all datasets. This indicates that ISM plays a critical role in extracting discriminative spectral features that complement the spatial modeling of OSM.

To directly compare ISM with a 1D-CNN baseline, we replaced ISM with a 1D-CNN of similar parameter count while keeping OSM unchanged. The proposed ISM consistently outperforms 1D-CNN on all five datasets, with improvements of +0.63% (PaviaU), +1.40% (WHU-HC), +0.87% (WHU-HH), and +1.05% (Houston 2013); on WHU-LK, the gain is marginal (+0.27%) but still positive. More importantly, ISM achieves lower or comparable variance, indicating better robustness.

The advantage of ISM stems from its selective state-space mechanism, which dynamically determines which spectral information to retain or discard across bands. Unlike 1D-CNNs with fixed receptive fields and static kernels, ISM can capture long-range, nonlocal spectral dependencies without increasing depth or using dilated convolutions. Simple attention mechanisms, while capable of global weighting, lack built-in sequential bias and may overfit to spectrally similar classes. In contrast, Mamba's recurrent-like state maintains a compressed history of previous bands, enabling more effective discrimination for classes with high interclass spectral similarity (e.g., WHU-HH dataset, which contains abundant mixed pixels). These results demonstrate that ISM and OSM operate collaboratively. ISM extracts fine-grained spectral features, OSM captures spatial context, and together they produce deeply fused spectral–spatial representations with superior stability and accuracy.

In addition, at an early stage of this work, we implemented a full patch version of ISM, where the module processes the spectral vectors of all pixels within a patch simultaneously.

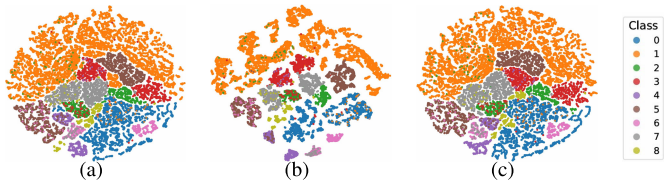


Fig. 14. t-SNE visualization of embedding features by ISM/OSM on PaviaU dataset. (a) First feature output by OSM. (b) Features output by ISM. (c) Second feature output by OSM.

However, this incurs prohibitive memory overhead (≈20 GB for a typical patch) due to maintaining hidden states for every pixel. With RCSS ensuring that the center pixel is highly representative of the patch, we instead process only the center pixel's spectrum. This reduces memory consumption dramatically while incurring less than 0.5% accuracy loss across all datasets. The spectral–spatial context of neighboring pixels is subsequently captured by the OSM.

2) Feature Visualization of ISM/OSM Blocks: In order to visually analyze the performance of ISM/OSM blocks, we employ the t-SNE algorithm to reduce the dimensionality of the high-dimensional features extracted by ISM/OSM blocks, enabling visualization in a 2-D space. The distributions of features on the PaviaU dataset extracted by ISM/OSM Mamba are illustrated in Fig. 14.

In the PaviaU dataset, the features produced by the ISM tend to be more compact, resulting in denser clusters and larger intercluster distances, while some fine-grained details are lost because spatial information is not explicitly modeled. And the features output by OSM the second time are more concentrated than those output the first time. For example, in the features from the first output, a small portion of samples for class 2 "Meadows" were not clustered, but in the subsequent second output, a complete cluster was formed. This is because the output of the OSM, after being enhanced by the output of ISM, embeds samples belonging to the same class into a more compact region of the feature space, thereby making the classification boundaries easier to identify.

F. Analysis on Different Patch Sizes

In HSIC tasks, image patches are commonly used as inputs to capture spectral information while fully exploiting spatial context. However, the selection of patch size has a significant impact on model performance. A patch that is too small may fail to effectively utilize spatial information, whereas an excessively large patch increases computational burden and introduces redundant information, ultimately degrading classification performance.

In our experiment, we evaluated OA, Params, and FLOPs for various patch sizes across five datasets. It is worth noting that, since our method requires a clearly defined center pixel, the patch size must be odd. Fig. 15 shows the performance comparison among five different patch sizes. The visualized results unmistakably showcase that alterations in patch size exert a significant influence on model performance. It can be observed that increasing the patch size from 7×7 to

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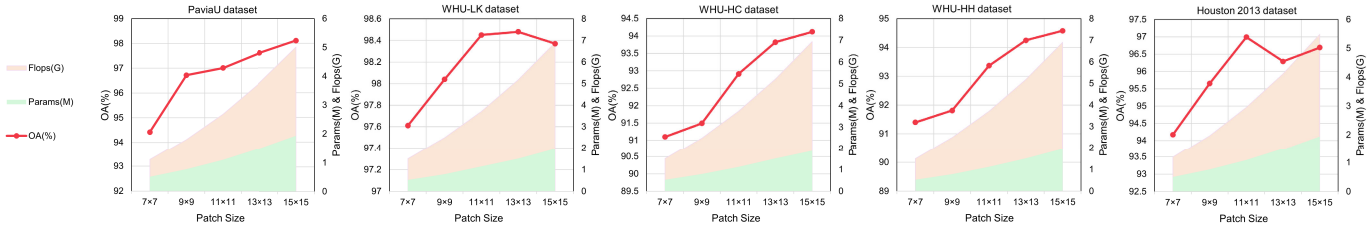


Fig. 15. OA, Params and FLOPs by using different patch sizes on five datasets.

TABLE VIII
QUANTITATIVE RESULT (ACC% ± STD%) OF THE PAVIAU DATASET

Class	CLOLN [11]	SSFTT [16]	GAHT [17]	GSC-ViT [18]	MambaHSI [26]	IGroupSS [27]	S2Mamba [31]	HSIMFormer [28]	Ours
Asphalt	**99.58±0.30**	92.55±2.51	92.07±3.22	95.22±1.83	93.99±2.46	94.76±2.30	94.86±2.11	92.97±2.52	96.49±1.38
Meadows	**98.07±0.66**	92.77±4.50	91.43±0.83	88.53±5.93	96.27±1.75	93.16±1.85	97.75±1.07	92.53±0.56	97.12±1.33
Gravel	93.00±3.44	92.19±4.59	**95.49±2.65**	94.93±3.07	93.27±5.50	93.25±1.65	90.51±2.29	85.31±4.29	91.96±4.45
Trees	96.75±1.45	97.61±1.43	96.11±2.67	94.63±2.47	89.95±3.89	95.60±1.59	96.27±1.24	92.86±2.22	**98.00±0.90**
Painted metal sheets	99.77±0.13	99.83±0.19	99.94±0.09	99.61±0.53	100.00±0.00	99.97±0.03	100.00±0.00	99.98±0.03	**100.00±0.00**
Bare Soil	92.25±6.11	98.27±1.42	96.68±1.10	97.90±1.82	98.59±0.97	**98.85±1.20**	96.20±1.55	97.04±2.66	95.56±6.49
Bitumen	87.51±5.93	96.00±4.25	99.19±0.28	**99.92±0.16**	96.14±3.19	99.66±0.59	99.53±0.28	99.80±0.84	99.77±0.16
Self-Blocking Bricks	92.29±1.57	91.89±3.08	90.66±1.91	94.00±3.02	94.13±3.17	93.51±2.94	95.38±2.44	93.32±3.86	**96.86±1.71**
Shadows	90.98±1.86	99.64±0.23	98.86±0.51	97.46±1.70	99.38±0.61	99.46±0.22	99.11±0.68	98.89±0.41	**99.67±0.13**
OA(%)	96.15±0.81	94.08±1.97	93.26±0.28	92.76±1.98	95.59±0.91	94.82±0.88	96.11±0.87	93.39±0.58	**97.01±1.76**
AA(%)	94.47±0.96	95.64±1.17	95.60±0.28	95.80±0.26	95.75±1.19	96.47±0.49	96.61±1.78	94.70±0.55	**97.58±1.53**
$\kappa \times 100$	94.89±1.05	92.26±2.50	91.19±0.36	90.60±2.47	95.19±1.63	93.28±1.14	95.52±1.59	91.42±0.72	**96.32±2.14**

11×11 leads to a significant improvement in OA across all five datasets. This is because the model can capture more useful spatial context, thereby enhancing classification accuracy. On the PaviaU dataset, the OA increased from 94.41% with a 7×7 patch to 97.01% with an 11×11 patch, while the number of parameters grew from 0.49 to 1.08 M and FLOPs increased from 1.09 to 2.70 G, which still remains within a manageable range. It is also observed that further enlarging the patch size continues to improve OA; with a 15×15 patch, OA reaches an impressive 98.11%. However, due to the corresponding doubling of computational cost, a patch size of 11×11 was chosen for subsequent experiments. Similarly, for the WHU-LK, WHU-HC, and WHU-HH datasets, after the patch size reaches 11×11, further increasing the patch size leads to a slower improvement in OA. For instance, for the WHU-HC dataset, the OA is 91.10% at a patch size of 7×7, which significantly increased to 92.91% at 11×11. Further increasing the patch size to 13×13 yielded an improvement to 93.82%. Notably, for the Houston 2013 dataset, the OA was 94.17% at a patch size of 7×7 and significantly improved to 96.79% at 11×11. However, further increasing the patch size to 13×13 led to a decrease in OA to 96.29%, accompanied by higher computational cost. This is because Houston 2013 is an urban scene with sparse labels and dense buildings; an excessively large patch size introduces too much redundant spatial information, which not only increases computational burden but also adversely affects the classification performance.

In summary, a patch size of 11×11 achieves an optimal balance across all datasets. It significantly improves classification accuracy while avoiding a drastic increase in computational cost and the potential redundant information associated with larger patch sizes. Therefore, we ultimately choose 11×11 as the patch size for our model input to ensure optimal performance and efficiency.

G. Classification Comparison With State-of-the-Art Models

In this study, we conducted a comprehensive comparison of our proposed L-HMambaS network with several state-of-the-art models. The selected baselines cover various model architectures, including CNN-based models [11], Transformer-based models [16], [17], [18], and Mamba-based model [26], [27], [28], [31]. Concretely, CLOLN [11] proposes a dual-single-channel 3-D convolution coupled with depth-wise convolution. SSFTT [16] introduces a spectral–spatial feature tokenization Transformer. GAHT [17] proposes a group-aware Transformer. GSC-ViT [18] incorporates a GSC module, combining GPWC and group convolution to enhance feature extraction. MambaHSI [26] stacks cascaded Mamba blocks with parallel spectral and spatial branches. IGroupSS [27] employs an IGSM mechanism that splits the spectral dimension into four regular groups to reduce spectral redundancy. S2Mamba [31] proposes a block-wise cross-scanning strategy to capture spatial correlations among pixels, complemented by a bidirectional spectral scanning strategy to capture interband relationships. HSIMFormer [28] further hybridizes Transformer and Mamba via an ITE-CME design.

The experiments were performed on five HSI datasets, with 30 training samples per class for each dataset. Tables VIII–XII present the OA, AA, κ, and accuracy per class for various models on five HSI datasets, highlighting the best results in bold. For a fair comparison, the patch size for all compared models and datasets was set to 11×11. For GSC-ViT [18], which adopts even patch sizes, a patch size of 12×12 is selected to minimize its potential impact on performance.

1) PaviaU Dataset: Table VIII provides a detailed quantitative comparison of various methods on the PaviaU dataset. Our method surpasses all competing models in terms of OA, AA, and Kappa coefficient. Specifically, the OA reaches

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TABLE IX
QUANTITATIVE RESULT (ACC% ± STD%) OF THE WHU-LK DATASET

Class	CLOLN [11]	SSFTT [16]	GAHT [17]	GSC-ViT [18]	MambaHSI [26]	IGroupSS [27]	S2Mamba [31]	HSIMFormer [28]	Ours
Corn	99.72±0.30	98.69±1.16	99.65±0.20	**99.74±0.37**	99.37±0.53	99.37±0.54	99.50±0.57	99.33±0.75	99.64±0.23
Cotton	99.50±0.62	97.77±1.90	97.25±1.43	97.72±1.43	97.97±1.60	98.61±0.92	99.49±0.39	94.63±5.52	**99.69±0.29**
Sesame	85.38±16.17	95.35±5.36	99.64±0.46	97.98±2.96	98.41±1.40	**99.82±0.10**	99.64±0.17	99.75±0.34	99.65±0.24
Broad-leaf soybean	**99.69±0.09**	96.95±1.98	95.09±3.01	93.17±2.02	93.85±3.06	95.16±2.56	96.04±0.71	95.02±1.92	96.62±0.83
Narrow-leaf soybean	80.58±6.88	97.03±3.54	**99.67±0.28**	95.89±2.96	98.37±1.23	98.90±0.72	95.76±2.11	98.53±0.87	98.16±1.22
Rice	99.62±0.20	98.57±0.95	99.08±0.60	98.03±2.04	98.84±1.43	98.24±2.71	97.83±0.76	99.78±0.22	**99.80±0.36**
Water	**99.85±0.12**	99.24±0.35	98.04±0.48	99.38±0.47	99.60±0.44	99.24±0.25	99.22±0.29	98.45±2.24	99.53±0.37
Roads and houses	93.44±1.25	97.24±1.63	93.13±2.21	93.99±2.76	93.59±1.08	95.06±1.34	**98.66±1.15**	96.26±0.52	97.14±1.67
Mixed weed	82.75±5.30	95.65±1.33	**97.05±2.12**	94.78±1.85	96.37±1.38	96.67±1.34	93.23±2.54	95.82±0.96	96.04±3.31
OA(%)	98.17±0.51	98.08±0.59	97.29±0.87	96.98±0.70	97.34±1.09	97.94±0.61	97.98±0.41	97.61±0.90	**98.45±0.30**
AA(%)	93.39±1.88	97.39±0.92	97.62±0.40	96.74±0.44	97.37±0.70	97.85±0.36	97.71±0.59	97.45±0.78	**98.44±0.33**
$\kappa \times 100$	97.60±0.66	97.48±0.77	96.46±1.12	96.06±0.90	96.63±1.86	97.30±0.77	97.35±0.48	96.89±1.10	**97.98±0.37**

TABLE X
QUANTITATIVE RESULT (ACC% ± STD%) OF THE WHU-HC DATASET

Class	CLOLN [11]	SSFTT [16]	GAHT [17]	GSC-ViT [18]	MambaHSI [26]	IGroupSS [27]	S2Mamba [31]	HSIMFormer [28]	Ours
Strawberry	**94.97±0.92**	82.54±6.08	93.44±2.30	92.95±2.90	89.48±5.65	88.86±1.85	89.53±2.12	90.62±3.60	92.22±2.15
Cowpea	**91.97±3.61**	85.26±4.14	88.26±4.93	84.51±4.80	81.82±6.16	80.73±2.01	82.67±1.88	79.87±1.72	86.19±1.31
Soybean	83.93±5.57	86.07±6.13	90.49±6.78	93.25±4.06	**94.91±3.69**	92.29±5.18	88.17±1.51	86.91±2.62	91.43±1.65
Sorghum	85.23±5.35	96.14±0.49	97.09±1.37	97.08±1.80	98.49±1.39	98.03±0.98	97.97±0.90	97.10±1.13	**98.51±0.76**
Water spinach	71.97±14.46	97.79±1.29	98.52±1.11	98.87±1.70	**99.55±0.86**	97.86±1.07	97.86±1.09	98.18±1.85	99.04±0.62
Watermelon	45.17±7.27	80.43±5.67	80.20±4.05	80.50±3.91	76.99±5.46	82.97±2.50	**86.65±3.14**	79.60±2.20	86.08±3.49
Greens	74.41±5.76	89.58±2.33	91.91±4.77	91.71±2.63	93.27±3.29	93.79±2.60	97.12±0.81	91.00±3.14	**97.18±0.78**
Trees	**82.93±4.22**	73.94±7.86	77.72±3.41	70.44±9.14	72.81±7.93	73.33±7.90	79.03±1.47	70.82±5.36	81.58±2.80
Grass	76.19±2.42	81.35±8.78	81.35±3.47	83.21±3.66	88.91±3.17	81.18±3.39	87.20±4.26	80.67±3.38	**88.96±2.79**
Red roof	92.91±8.27	93.70±2.71	97.71±0.59	95.30±4.58	94.41±1.86	96.43±2.74	97.49±0.93	96.25±2.09	**98.04±0.85**
Gray roof	87.37±7.16	87.31±4.75	91.04±4.90	91.35±2.72	90.77±4.28	96.10±0.93	96.90±0.73	95.02±1.06	**98.26±0.57**
Plastic	73.41±10.73	88.74±6.37	94.38±2.72	94.66±2.37	90.36±5.71	93.87±3.50	96.39±2.77	90.24±2.79	**98.05±2.63**
Bare soil	68.51±4.74	71.76±6.68	81.08±6.17	80.33±11.56	79.24±4.88	78.75±4.53	81.57±4.18	72.27±4.46	**84.44±2.23**
Road	**90.77±3.91**	79.83±4.77	84.30±2.46	80.65±2.64	89.70±3.37	79.62±4.28	85.44±2.49	79.61±3.45	88.10±2.54
Bright object	74.67±10.40	97.37±1.90	96.30±3.34	93.57±3.99	94.19±2.64	90.78±1.46	98.78±1.76	89.15±5.80	**98.86±0.61**
Water	**99.39±0.96**	96.30±2.25	97.64±1.45	97.87±1.61	98.40±0.90	97.76±1.25	99.05±1.12	94.27±6.55	98.34±0.47
OA(%)	89.03±1.12	87.13±1.07	91.33±0.56	90.28±0.57	90.52±1.44	89.51±0.76	91.30±0.62	88.30±1.71	**92.91±0.28**
AA(%)	80.86±2.33	86.76±0.62	90.09±0.36	89.14±0.98	89.58±1.23	88.93±0.37	91.35±1.51	86.99±0.60	**92.80±0.35**
$\kappa \times 100$	87.21±1.29	85.05±1.20	89.89±0.63	88.68±0.66	88.97±2.22	87.49±0.92	89.87±1.08	86.06±1.85	**91.67±0.30**

TABLE XI
QUANTITATIVE RESULT (ACC% ± STD%) OF THE WHU-HH DATASET

Class	CLOLN [11]	SSFTT [16]	GAHT [17]	GSC-ViT [18]	MambaHSI [26]	IGroupSS [27]	S2Mamba [31]	HSIMFormer [28]	Ours
Red roof	94.68±4.09	91.03±5.11	94.72±1.68	90.67±5.89	95.22±2.64	**95.91±1.09**	95.87±2.17	92.49±5.14	95.35±1.70
Road	75.76±5.87	93.17±2.35	94.51±1.21	91.70±5.98	93.04±4.51	92.95±4.17	95.35±1.26	92.15±3.83	**95.62±1.10**
Bare soil	**97.63±1.65**	84.01±7.30	84.90±2.04	86.60±3.06	92.26±2.21	86.99±2.57	91.86±3.14	88.07±1.90	91.68±3.67
Cotton	**99.67±0.14**	92.63±2.14	96.10±1.33	96.76±0.48	95.47±2.66	93.37±2.96	96.81±2.15	93.31±1.66	96.62±0.78
Cotton firewood	65.68±15.46	94.89±2.49	95.65±2.69	94.28±1.04	95.08±1.67	93.96±2.52	**96.60±2.78**	93.12±1.81	95.01±3.14
Rape	**96.74±1.93**	91.11±1.37	92.38±2.24	92.86±3.41	92.43±2.06	92.36±4.36	90.33±1.74	93.36±0.81	93.79±2.50
Chinese cabbage	**86.90±3.95**	75.93±4.80	81.50±2.46	79.89±5.25	81.80±4.71	80.22±4.24	83.69±2.87	72.27±3.48	82.27±3.88
Pakchoi	47.61±10.32	79.67±7.67	75.75±5.79	75.82±6.57	92.82±3.73	79.99±5.12	85.34±4.18	82.40±5.72	**92.88±3.58**
Cabbage	**98.91±0.89**	95.75±2.73	96.15±2.05	95.91±3.21	92.65±1.98	96.63±1.74	96.40±1.57	94.91±0.85	98.21±1.77
Tuber mustard	89.90±4.88	91.53±3.10	90.33±2.89	**92.43±1.76**	85.54±4.67	82.13±1.68	87.21±2.13	83.09±3.60	87.27±3.99
Brassica parachinensi	74.88±11.82	71.55±11.14	82.22±5.45	82.54±5.47	87.42±5.47	82.93±3.07	80.14±4.13	82.47±2.20	**90.48±3.50**
Brassica chinensis	75.18±6.84	80.95±4.29	76.60±2.30	81.92±4.65	**82.35±6.60**	78.79±7.02	78.25±3.67	73.01±2.55	80.97±4.25
Small Brassica chiner	84.30±4.22	78.89±3.16	80.38±3.23	74.80±4.95	82.82±2.48	73.37±2.65	**85.65±3.03**	68.69±2.76	83.77±3.97
Lactuca sativa	86.80±8.33	95.10±2.48	95.75±2.30	95.24±2.67	92.02±1.55	92.81±2.57	**96.21±1.67**	90.00±3.77	95.88±1.07
Celtuce	70.06±11.15	98.03±1.14	99.33±0.63	**99.64±0.25**	96.93±3.15	97.06±2.35	97.79±2.05	96.89±1.36	98.58±1.01
Film covered lettuce	97.41±2.47	97.11±1.39	96.64±2.79	97.70±2.50	96.30±1.96	92.30±1.37	97.32±1.94	90.64±2.28	**97.63±1.75**
Romaine lettuce	87.06±8.35	95.39±4.32	97.06±1.68	95.84±2.80	94.93±5.81	95.04±1.59	96.99±2.16	95.63±1.19	**97.10±1.20**
Carrot	66.61±6.88	98.21±0.75	96.68±0.76	97.15±1.37	94.23±3.80	97.60±1.09	98.39±0.85	**98.81±0.93**	98.39±0.80
White radish	84.85±11.92	90.34±3.79	92.51±2.02	91.91±3.54	90.44±2.94	91.43±2.34	89.09±1.96	92.82±1.39	**93.16±2.35**
Garlic sprout	75.68±9.72	93.38±2.51	95.27±1.38	95.73±1.88	**98.68±0.78**	95.65±2.57	97.44±1.58	93.64±2.10	97.77±0.45
Broad bean	30.13±7.59	93.65±5.08	95.48±1.32	88.44±7.25	99.57±0.67	**99.58±0.44**	99.46±0.55	99.20±0.76	99.44±0.79
Tree	47.65±11.85	95.68±4.11	96.81±1.86	97.80±2.22	98.24±2.64	94.74±4.19	97.12±1.49	95.14±2.58	**98.45±1.62**
OA(%)	89.78±0.88	89.36±0.79	91.84±0.63	91.83±0.44	92.26±1.07	90.23±1.07	92.50±1.28	89.17±0.84	**93.37±0.62**
AA(%)	78.82±0.80	89.91±0.75	91.22±0.71	90.71±0.42	92.29±0.86	90.38±0.47	92.34±0.91	89.39±0.58	**93.67±0.52**
$\kappa \times 100$	87.23±1.03	86.78±0.93	89.78±0.77	89.75±0.55	91.90±1.86	87.83±1.22	90.84±1.14	86.69±0.90	**91.55±0.89**

97.01% ± 1.76%, exceeding the second-best model (CLOLN [11]) by 1 percentage points, while the AA also achieves 97.58% ± 1.53%, outperforming the second-best model (IGoupSS [27]) by 1.11 percentage points. It is worth noting that for the single-class recognition of *Self-Blocking Bricks*, our method achieves an accuracy of 96.86%.

As visually substantiated in Fig. 16, our method exhibits clearer class boundaries, especially within the white box shown in the figure, where *Asphalt* and *Self-Blocking Bricks* display elongated spatial distributions. Other methods perform less effectively in this region, presenting blurred boundaries or a loss of the original spatial distribution patterns. The boundary

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TABLE XII
QUANTITATIVE RESULT (ACC% ± STD%) OF THE HOUSTON 2013 DATASET

Class	CLOLN [11]	SSFTT [16]	GAHT [17]	GSC-ViT [18]	MambaHSI [26]	IGroupSS [27]	S2Mamba [31]	HSIMFormer [28]	Ours
Healthy grass	95.03±3.63	92.96±5.99	94.12±5.12	91.19±5.09	94.91±4.56	97.73±0.99	98.41±2.74	93.70±2.15	**98.98±0.41**
Stressed grass	86.81±8.64	95.83±2.59	94.52±3.23	96.83±2.38	97.77±1.48	98.48±0.77	98.07±1.76	**98.69±1.09**	98.24±0.54
Synthetic grass	98.67±3.98	99.91±0.12	99.75±0.25	98.83±1.26	99.77±0.33	99.61±0.47	100.00±0.00	99.57±0.70	**100.00±0.00**
Trees	95.71±3.02	95.98±2.16	93.90±2.46	92.13±4.24	97.19±2.26	**98.94±1.17**	92.35±3.19	98.93±0.60	98.35±1.97
Soil	96.86±3.01	99.45±0.56	99.65±0.58	100.00±0.00	99.73±0.39	99.98±0.03	100.00±0.00	99.85±0.30	**100.00±0.00**
Water	94.05±3.56	98.33±1.76	98.40±1.62	99.20±0.84	97.78±2.40	99.59±0.81	99.67±0.33	99.12±1.05	**100.00±0.00**
Residential	81.93±5.21	89.74±3.93	92.53±4.05	91.41±2.88	93.48±3.02	94.05±3.66	99.03±0.97	92.21±4.03	**99.06±0.32**
Commercial	**98.09±2.13**	81.04±8.61	83.52±5.43	82.73±4.22	79.67±3.39	85.01±5.30	91.11±2.39	81.73±8.58	90.33±2.91
Road	88.85±4.58	88.20±2.38	87.54±4.65	88.69±5.25	89.26±3.42	87.20±3.59	89.19±3.21	86.69±5.29	**89.42±3.06**
Highway	77.59±5.02	92.81±5.70	93.51±5.57	96.67±2.01	96.21±2.07	**99.55±0.33**	98.69±1.69	98.16±2.56	98.70±1.53
Railway	88.92±6.29	89.40±4.87	90.35±3.85	92.41±3.45	90.86±2.89	98.92±1.17	89.52±2.85	**99.14±0.74**	95.54±1.82
Parking Lot 1	82.30±7.23	92.80±3.38	91.14±2.44	91.63±2.79	91.37±3.41	**96.11±2.45**	91.80±1.45	94.03±4.22	94.35±2.74
Parking Lot 2	90.56±2.66	96.28±1.79	96.28±1.66	95.80±1.74	97.88±2.50	95.19±3.32	98.79±1.32	97.86±1.66	**99.05±0.64**
Tennis Court	98.80±1.78	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	99.95±0.10	99.74±0.31	99.95±0.10	**100.00±0.00**
Running Track	97.79±1.00	99.97±0.07	99.67±0.40	99.48±0.89	96.31±2.69	**100.00±0.00**	99.67±0.64	99.97±0.06	99.67±0.57
OA(%)	89.51±1.13	93.00±1.16	93.20±1.02	93.39±1.27	93.91±0.81	96.27±0.72	95.05±1.06	94.92±1.62	**96.79±0.57**
AA(%)	91.46±0.89	94.18±1.02	94.32±0.87	94.47±0.99	94.81±0.80	96.67±0.76	95.98±1.37	95.88±1.35	**97.36±0.48**
$\kappa \times 100$	88.65±1.22	92.43±1.26	92.65±1.10	92.85±1.37	94.07±1.75	95.94±0.80	94.65±1.69	94.47±1.77	**96.68±0.51**

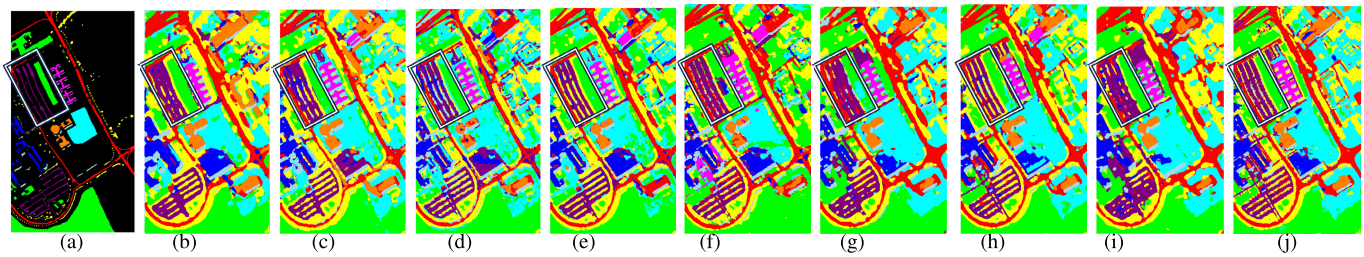


Fig. 16. Classification maps obtained by different models for the PaviaU dataset. These maps correspond to a single experiment in Table VIII. (a) GT. (b) CLOLN. (c) SSFTT. (d) GAHT. (e) GSC-ViT. (f) MambaHSI. (g) IGroupSS. (h) S2Mamba. (i) HSIMFormer. (j) Ours.

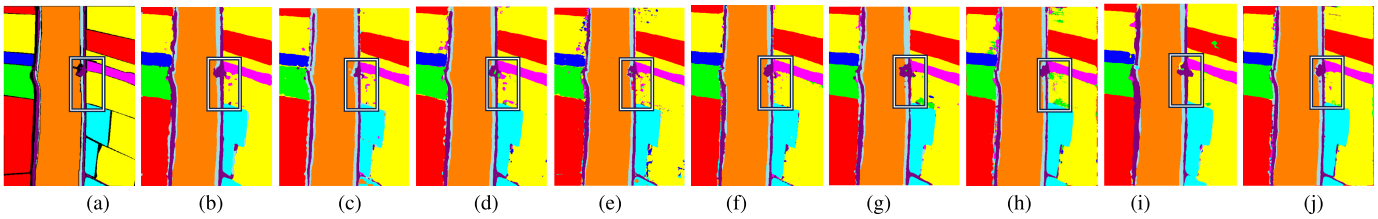


Fig. 17. Classification maps obtained by different models for the WHU-LK dataset. These maps correspond to a single experiment in Table IX. (a) GT. (b) CLOLN. (c) SSFTT. (d) GAHT. (e) GSC-ViT. (f) MambaHSI. (g) IGroupSS. (h) S2Mamba. (i) HSIMFormer. (j) Ours.

ambiguity mainly arises from the confusion between spectrally similar categories, while the overlap of spatial samples further contributes to the degradation of the original spatial structure. In contrast, our method maintains sharp boundaries and preserves the inherent spatial distribution, demonstrating its superiority in handling spatial overlaps and deeply exploiting spectral–spatial correlations.

2) WHU-Hi-LongKou Dataset: As quantitatively demonstrated in Table IX, for this dataset, most models can achieve satisfactory results, but our method achieves state-of-the-art results, leading in OA (98.45%) and AA (98.44%) with minimal standard deviation, showcasing its stability and reliability. For example, as shown in the white box in Fig. 17, which marks a compact region where five land-cover types are adjacent, our method achieves the best inference performance with clear and well-defined boundaries. In particular, for the category of Roads and Houses, other models tend to produce blurred boundaries or misclassify parts of the Broad-leaf Soybean (yellow regions) as Roads and Houses. In contrast, our method maintains high accuracy in these regions, effectively distinguishing between similar categories.

3) WHU-Hi-HanChuan Dataset: The proposed method achieves superior performance on the WHU-HC dataset across three key evaluation metrics, reaching state-of-the-art results. In Table X, our model achieves an OA of 92.91% ± 0.28%, an AA of 92.80% ± 0.35%, and a Kappa coefficient of 91.67% ± 0.30%, exceeding the second-best competing model GAHT [17] by 1.58%, 2.71%, and 1.78%, respectively. Particularly noteworthy is that in terms of per-class recognition accuracy, our model outperforms others in eight leading categories, with the recognition accuracies of *watermelon* (86.08%) and *greens* (97.18%) exceeding those of other models by more than 4%. For the second-best competing model GAHT [17], it fuses local–global features through grouped pixel embedding (GPE) and hierarchical transformer structures; its fixed-window attention mechanism struggles to adapt to complex spatial distributions. L-HMambaS, however, by introducing an RCSS and more effectively capturing the

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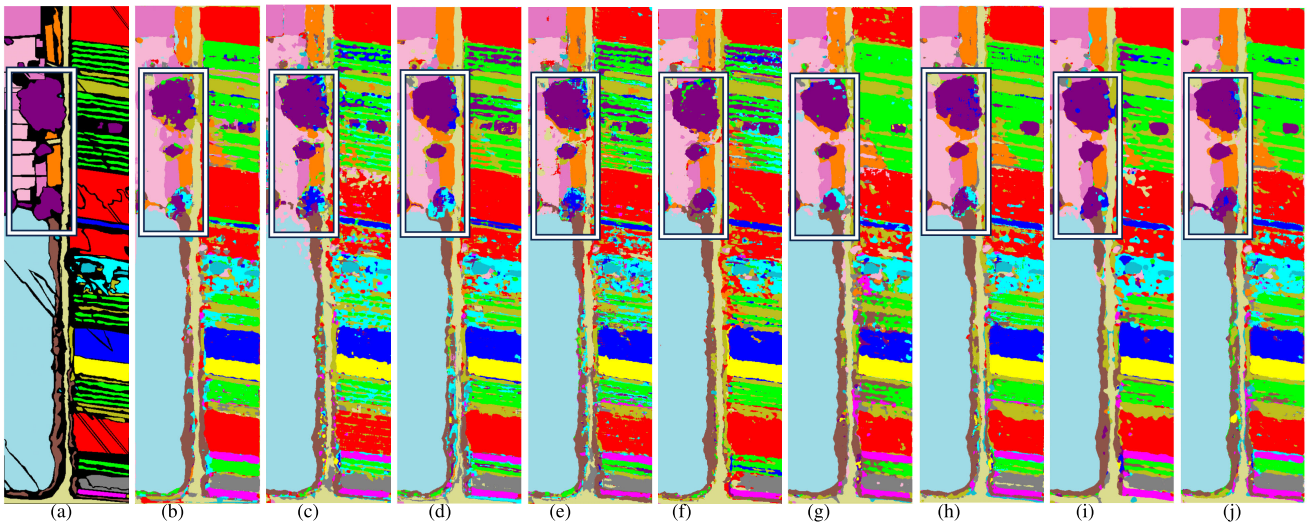


Fig. 18. Classification maps obtained by different models for the WHU-HC dataset. These maps correspond to a single experiment in Table X. (a) GT. (b) CLOLN. (c) SSFTT. (d) GAHT. (e) GSC-ViT. (f) MambaHSI. (g) IGroupSS. (h) S2Mamba. (i) HSIMFormer. (j) Ours.

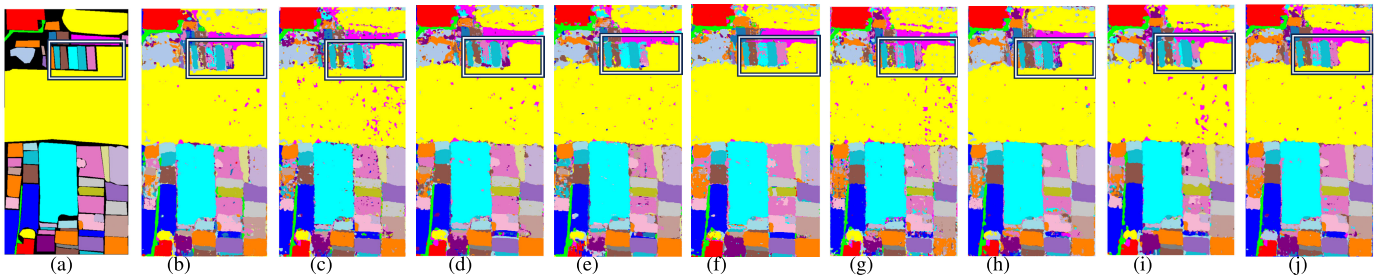


Fig. 19. Classification maps obtained by different models for the WHU-HH dataset. These maps correspond to a single experiment in Table XI. (a) GT. (b) CLOLN. (c) SSFTT. (d) GAHT. (e) GSC-ViT. (f) MambaHSI. (g) IGroupSS. (h) S2Mamba. (i) HSIMFormer. (j) Ours.

intrinsic correlations between spectral and spatial information, exhibits superior robustness in complex scenes and reveals deeper data-level dependencies. As shown in the white box in Fig. 18, other models have lower recognition accuracy for the tree and its surrounding categories compared to the proposed model.

4) WHU-Hi-HongHu Dataset: For the WHU-HH dataset, as quantitatively demonstrated in Table XI, our method attains an OA of 93.37% $\pm$ 0.62%, and AA of 93.67% $\pm$ 0.52%. The study area of the WHU-HH dataset represents a complex agricultural scene containing various crop categories. Moreover, this region includes different varieties of the same crop, such as Chinese cabbage and cabbage, which results in high spectral similarity between classes. Furthermore, since the dataset simulates a fragmented land pattern, a large proportion of mixed pixels exists in boundary regions, which amplifies interclass similarity and intraclass variability. Benefiting from the RCSS in L-HMambaS, the impact of fragmented land patterns and mixed pixels is effectively reduced as shown in the white box in Fig. 19. For cases with high interclass spectral similarity, the ISM enables more fine-grained spectral feature learning, while the combination of inner and OSM allows deeper exploration of the latent spectral–spatial semantics. Consequently, our method exhibits strong stability in large-scale multiclass datasets.

5) Houston 2013 Dataset: On the Houston 2013 dataset, as quantitatively demonstrated in Table XII, our model maintained top performance with an OA of 96.79% $\pm$ 0.57%, AA of 97.36% $\pm$ 0.48%, and Kappa coefficient of 96.68% $\pm$ 0.51%, and IGroupSS [27] achieves the second-best accuracy. IGroupSS [27] divides the high-dimensional spectral features of HSI into four nonoverlapping groups at fixed intervals and models each group using unidirectional S6 scanning in four directions, enabling multiscale global spectral–spatial feature extraction. However, its intergroup interaction is limited, relying solely on channel attention for connection, which constrains deep semantic fusion across groups. In contrast, our method exploits the spatial homogeneity of pixels and deeply explores the intrinsic relationship between spectral and spatial dimensions to achieve comprehensive semantic integration. As shown in Fig. 20, the material boundaries within the white boxes are clear, and the shapes of buildings and roads remain intact. This indicates that L-HMambaS not only achieves higher classification accuracy but also provides superior full-scene inference under sparse labeling conditions.

6) Classification Performance Under Different Numbers of Training Samples: As quantitatively demonstrated in Table XIII, the classification performance of the proposed method and comparison method is compared under different numbers of training samples. We conduct comparative experiments on three benchmark datasets, including PaviaU, WHU-HC, and Houston 2013. Specifically, we randomly selected 30, 40, 50, 60, and 70 labeled samples per class for training with the remaining used for testing, repeating

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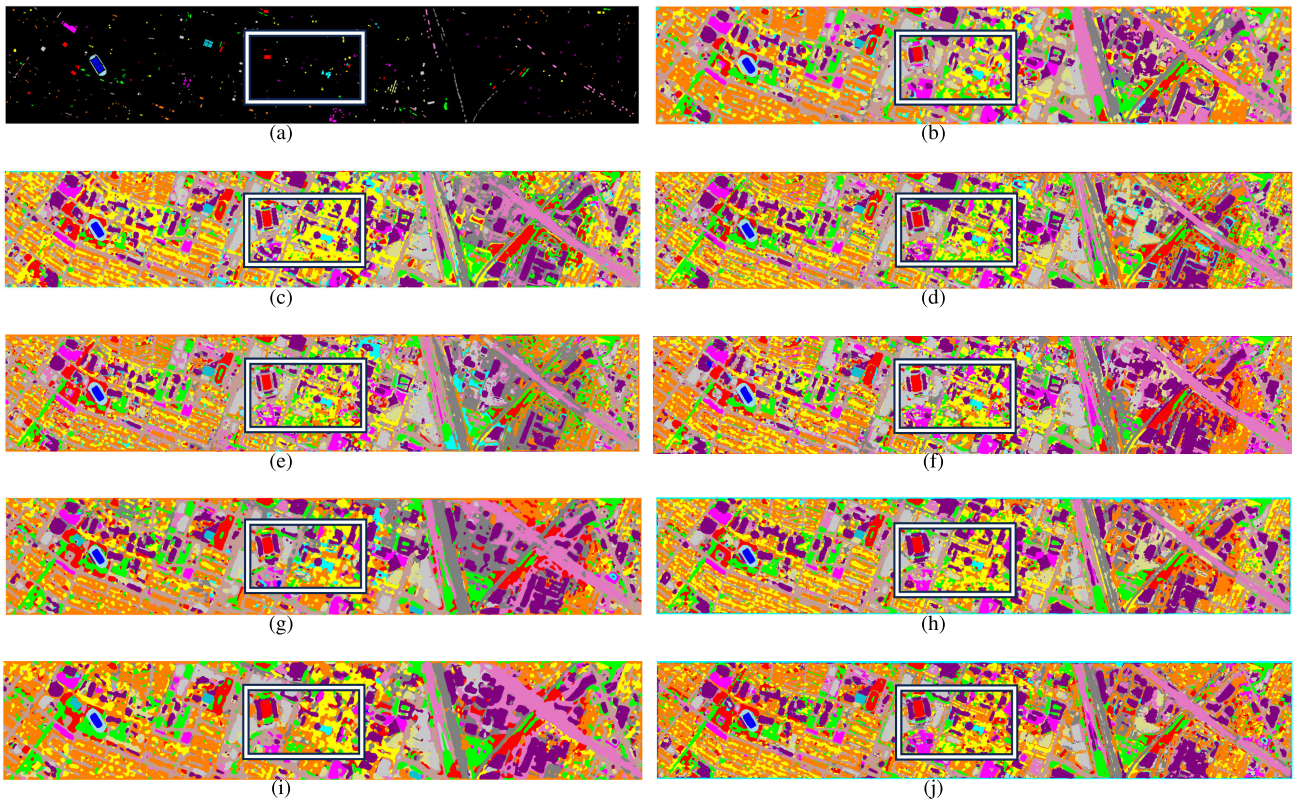


Fig. 20. Classification maps obtained by different models for the Houston 2013 dataset. These maps correspond to a single experiment in Table XII. (a) GT. (b) CLOLN. (c) SSFTT. (d) GAHT. (e) GSC-ViT. (f) MambaHSI. (g) IGroupSS. (h) S2Mamba. (i) HSIMFormer. (j) Ours.

TABLE XIII
QUANTITATIVE RESULT (ACC% ± STD%) OF THE THREE DATASETS UNDER DIFFERENT NUMBERS OF TRAINING SAMPLES

Dataset	Quantity	CLOLN [11]	SSFTT [16]	GAHT [17]	GSC-ViT [18]	MambaHSI [26]	IGroupSS [27]	S2Mamba [31]	HSIMFormer [28]	Ours
PaviaU(OA[%])	30 per class	96.15±0.81	94.08±1.97	93.26±0.28	92.76±1.98	95.59±0.91	94.82±0.88	93.74±1.29	93.39±0.58	**97.01±1.76**
	40 per class	95.87±1.30	95.94±1.29	95.65±0.52	96.37±1.12	96.93±0.83	94.92±2.83	95.22±1.27	95.22±0.65	**97.82±0.62**
	50 per class	93.00±2.32	96.89±2.18	96.90±0.76	96.93±0.60	97.67±0.64	97.42±1.09	96.85±1.31	96.36±0.90	**98.18±1.99**
	60 per class	97.63±0.56	97.81±1.01	97.34±0.60	97.27±0.87	98.03±0.50	96.16±2.63	97.83±0.94	97.21±0.31	**98.63±0.57**
	70 per class	98.00±0.35	98.64±0.20	97.84±0.38	97.79±0.85	98.26±0.33	97.81±1.02	98.21±0.75	97.45±0.83	**98.78±0.61**
WHU-HC(OA[%])	30 per class	89.03±1.12	87.13±1.07	91.33±0.56	90.28±0.57	90.52±1.44	89.51±0.76	88.21±2.19	88.30±1.71	**92.91±0.28**
	40 per class	88.47±1.45	88.29±1.58	92.31±0.29	92.01±0.81	92.99±0.72	89.96±1.41	91.22±1.31	90.68±0.27	**94.06±0.65**
	50 per class	89.58±1.18	89.68±0.81	92.27±1.02	92.10±0.65	94.34±0.76	88.03±5.83	93.81±0.72	91.21±0.69	**94.48±0.39**
	60 per class	90.64±0.95	90.72±0.61	93.76±0.73	94.10±0.30	95.14±0.82	91.94±2.01	94.98±0.63	91.25±1.42	**95.48±0.58**
	70 per class	90.69±1.45	91.59±1.05	94.74±0.54	93.66±0.50	95.15±0.74	92.26±0.65	95.03±0.57	91.82±0.51	**95.80±0.77**
HS 2013(OA[%])	30 per class	89.51±1.13	93.00±1.16	93.20±1.02	93.39±1.27	93.91±0.81	96.27±0.72	93.97±1.63	94.92±1.62	**96.79±0.57**
	40 per class	94.38±0.81	94.80±0.71	94.99±0.35	95.65±0.16	95.11±0.82	96.65±0.57	95.22±1.29	96.00±0.26	**96.96±0.62**
	50 per class	96.24±0.38	96.38±0.75	96.70±0.14	96.30±0.55	96.38±0.49	97.41±0.29	96.21±1.07	97.36±0.53	**97.46±0.52**
	60 per class	96.06±0.49	92.02±0.57	94.15±0.64	95.02±0.39	97.05±0.23	98.01±0.60	97.31±0.31	97.86±0.64	**98.16±1.10**
	70 per class	97.37±0.49	96.42±0.82	96.94±0.11	96.41±0.48	97.41±0.32	98.55±0.26	98.04±0.39	97.91±0.48	**98.69±0.35**

each experiment five times and reporting the average OA. The L-HMambaS consistently achieves the highest OA in all 15 experimental settings. Notably, the proposed method demonstrates particularly remarkable performance in low-data regimes (30 and 40 samples per class), where HSIC is notoriously challenging due to limited supervision. For instance, the proposed method achieves gains of up to 1.58% over the second-best competitor GAHT on WHU-HC with 30 samples per class, with consistent advantages of 1.15% on PaviaU and 0.41%–0.52% on HS 2013. This superiority stems from the method's enhanced capability to effectively capture and exploit intricate spectral–spatial dependencies, mitigating overfitting and maximizing informational yield from scarce labeled examples. As the number of training samples increases from 50 to 70 per class, OA improvements across all compared methods become progressively incremental and saturated, reflecting the diminishing returns typical in well-explored hyperspectral benchmarks. Nevertheless, the proposed approach steadfastly maintains its lead in every configuration, with margins ranging from 0.23% to 0.92% averaged across datasets. This sustained dominance underscores its exceptional effectiveness in leveraging spectral–spatial features even under severe annotation constraints.

In summary, our L-HMambaS network demonstrates superior accuracy, stability, and robustness across five datasets and diverse scenarios, reinforcing its advancement and broad applicability in HSIC tasks. For example, on the Houston 2013 dataset, which represents a complex urban environment with high spectral variability, hierarchical Mamba exhibits outstanding capability in jointly modeling spectral discrepancies and

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TABLE XIV
PERFORMANCE COMPARISON OF NINE MODELS ON PAVIAU DATASET

	Model	Params/K	FLOPs/G	Train Time/s	Test Time/s
PU	CLOLN [11]	**3.53**	**0.0147**	**46.47**	14.35
	SSFTT [16]	484.42	0.07	68.95	7.73
	GAHT [17]	927.11	14.35	119.75	16.17
	GSC-VIT [18]	130.32	1.43	128.89	16.77
	MambaHSI [26]	412.87	39.18	295.76	**0.04**
	IGroupSS [27]	57.31	0.40	84.51	108.49
	S2Mamba [31]	120.51	0.84	61.54	9.53
	HSIMFormer [28]	170.47	1.37	53.75	34.85
	Ours	1082.58	2.70	82.06	15.84

spatial configurations, enabling the model to maintain high accuracy. These results collectively highlight the proposed method's ability to balance precision and reliability in complex hyperspectral scenarios.

H. Analysis on Computational Complexity

Table XIV presents the comparison of nine models on the PaviaU dataset in terms of Params (K), FLOPs (G), train time, and test time, where test time denotes the time required to test a whole hyperspectral image. Although L-HMambaS introduces a moderate increase in parameter count (1.08 M) compared to other models, it maintains an efficient inference time of 15.84 s on the PaviaU dataset. Meanwhile, its FLOPs reach 2.70 G, which remains within a reasonable range and is notably lower than 39.18 G of MambaHSI [26] and 14.35 G of GAHT [17], but still considerably higher than CLOLN's 0.0147 G [11] and SSFTT [16]'s 0.07 G. In terms of training time, since our model employs a nested Mamba structure, the hierarchical Mamba block has a time complexity of $\mathcal{O}(n^2)$. However, due to efficient handling of the input data at the ISM, the training time is still controlled within a reasonable range of 82.06 s. Meanwhile, the full-image inference speed of 15.84 s is also within the average range of similar models, faster than HSIMFormer's 34.85 s [28], and significantly faster than IGroupSS's 108.49 s [27]. Overall, our model achieves a significant accuracy advantage over previous models, accompanied by increased computational cost; however, the two are well balanced, demonstrating substantial potential in real-world HSIC tasks.

V. CONCLUSION

This study proposes an advanced HSIC method, termed L-HMambaS, which is inspired by linguistic structures and incorporates an RCSS strategy. Across five HSI datasets, our model achieves the highest OA, demonstrating superior classification capability and strong generalization ability, even under limited training samples. Ablation studies confirm that the use of RCSS provides significant advantages in eliminating spatial overlap and mitigating the impact of boundary-mixed pixels. They also verify the critical role of the ISM, which, when combined with the OSM, effectively captures deeply fused spectral–spatial semantic features. Through extensive experiments, we identified the optimal configuration that balances classification accuracy and computational efficiency. Overall, L-HMambaS demonstrates outstanding performance in HSIC tasks, particularly in complex agricultural and urban scenes. Its capability to extract deeply fused spectral–spatial features enables excellent generalization during full-image inference, resulting in clear class boundaries and well-preserved object shapes. However, we acknowledge that certain limitations remain in terms of computational efficiency and performance under few-shot learning scenarios. Future research can explore the integration of cross-domain learning paradigms with advanced few-shot classification techniques to enhance the model's adaptability and robustness across diverse application scenarios while reducing computational overhead.

ACKNOWLEDGMENT

The authors would like to take this opportunity to thank the editors and the anonymous reviewers for their detailed comments and suggestions, which greatly helped them to improve the clarity and presentation of their manuscript.

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